Preliminary GEK-85739B



GE Drive Systems

INSTRUCTIONS

# ECMS

## Extended Crane Management System

*These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.*

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# 1. SAFETY PRECAUTIONS

## WARNING, CAUTION AND NOTE LABELS PLACED ON THE EQUIPMENT

The following format is used on the safety and informative labels placed on the equipment. Read all labels and follow the directions of them whenever working on the equipment.

**WARNING:** Denotes operating procedures and practices that may result in personal injury or loss of life if not correctly followed.
WARNING labels will be red in color with black or white lettering.

**CAUTION:** Denotes operating procedures and practices that, if not strictly observed, may result in damage to, or destruction of the equipment.
CAUTION labels will be amber in color with black lettering.

**NOTE:** Notes call attention to information that is especially significant in understanding and operating the equipment.
NOTE labels will be white in color with black lettering.

## WARNING, CAUTION AND NOTE PARAGRAPHS WITHIN THIS INSTRUCTION

The following paragraphs list some general safety reminders and safety recommendations to be followed when installing or operating this equipment. These safety precautions will be repeated throughout this instruction book where applicable.

**WARNING - ELECTRICAL SHOCK OR INJURY FROM ROTATING EQUIPMENT DUE TO REMOTE ENERGIZATION OR OPERATION OF EQUIPMENT HAZARD: No modification to the equipment or software should be attempted without first reviewing the effect on the overall system and taking the necessary precautions to assure safety of personnel.**

**WARNING - ELECTRICAL SHOCK OR INJURY FROM ROTATING EQUIPMENT DUE TO REMOTE ENERGIZATION OR OPERATION OF EQUIPMENT HAZARD: Assure safety of personnel throughout the entire system before operating this and associated equipment.**

**WARNING - ELECTRICAL SHOCK AND ROTATING EQUIPMENT INJURY HAZARD: Read and observe all safety precautions listed in the drive controller Instruction Book (GEK) when operating or troubleshooting this equipment.**

**CAUTION:** Do not remove connections or printed circuit cards while power is applied. This can damage the equipment.

**CAUTION:** Installation wiring must be in accordance with the National Electrical Code and be consistent with all local codes.

**NOTE:** *Always read the complete instructions prior to applying power or troubleshooting the equipment and follow the start-up procedures step by step.*

**NOTE:** *Read and heed all **WARNING, CAUTION** and **NOTE** labels posted on the equipment.*

# NOTES:

# 2. RECEIVING AND STORAGE GUIDELINES

## RECEIVING

The equipment and software should be placed under adequate cover immediately upon receipt as packing cases are not suitable for outdoor or unprotected storage. Each shipment should be carefully examined upon arrival and checked with the packing list. Any shortage or damage should be reported promptly to the carrier. If required, assistance may be requested from **GE DRIVE SYSTEMS, ERIE, PA.** When seeking assistance, please use **serial number, requisition number, model number** and **drive code** to identify the equipment **(Telephone 814-875-3851).**

## STORAGE

**Hardware:**

If this equipment is not to be installed immediately, hardware should be stored in a clean, dry location at ambient temperatures from -20° C (-4° F) to 55° C (131° F). The surrounding air must be free of chemical and electrically conductive or corrosive contaminants.

Precautions should be taken to prevent condensation from forming within the equipment enclosure. If the storage enviroment exceeds a 15° C (27° F) drop in temperature at 50% humidity over a four hour period, a heat source should be provided to prevent condensation. (A 100 watt lamp can sometimes serve as a source of heat). Higher humidities with smaller temperature changes can also cause condensation.

**Software:**

Software provided on magnetic media, such as floppy disks and magnetic tape, should be stored in a clean, dry location at ambient temperatures of 10° C to 55° C (50° F to 125° F). Care should be taken to protect the media from magnetic fields, high heat or humidity and physical damage.

# *NOTES:*

# 3. INTRODUCTION

The **Extended Crane Management System (ECMS)** is a combination of sophisticated hardware and advanced multi-tasking software. When coupled to the already powerful combination of the GE Series Six™ Programmable Controller, the GE Genius™ I/O System, and a GE Drive System, **ECMS** becomes a powerful monitoring and diagnostic tool.

The **ECMS** provides a "Window" into the electrical system including DC drives, AC motor control, AC supply system, limit switches, solenoids, and operator control. The **ECMS** hardware is comprised of an industrial computer mounted in a shockproof enclosure together with a VGA color monitor, an industrial grade keyboard, and when specified, a dot-matrix printer.

The **ECMS** uses a high speed parallel link to the Series Six Programmable Controller from an extension board located in the computer. The computer runs under a multi-tasking operating system called **Concurrent DOS .**

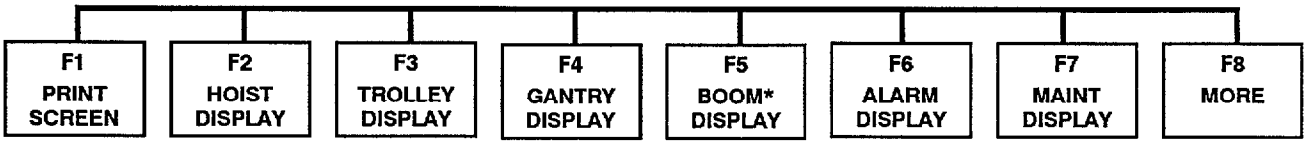
*TM - Series Six is a trademark of GE Fanuc*
*TM - Genius is a trademark of General Electric Company*

## SCREEN LEVELS

### TITLE SCREEN

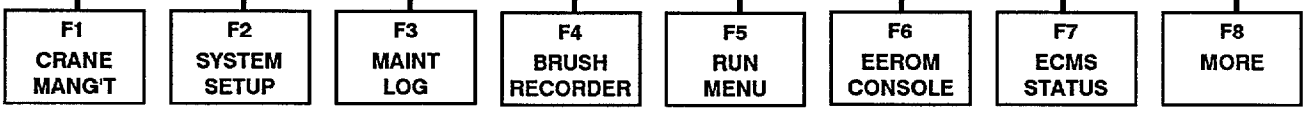



### SYSTEM SCREEN




**Or "WHEEL TURNING"; see seperate screen example.*

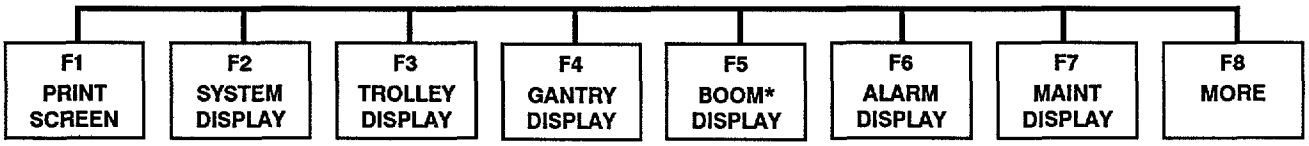
### MORE LEVEL*




**This "MORE LEVEL" is accessible from the* **TITLE, SYSTEM, HOIST, TROLLEY, GANTRY, BOOM,** *and* **WHEEL TURNING** *(if provided) Screens.*

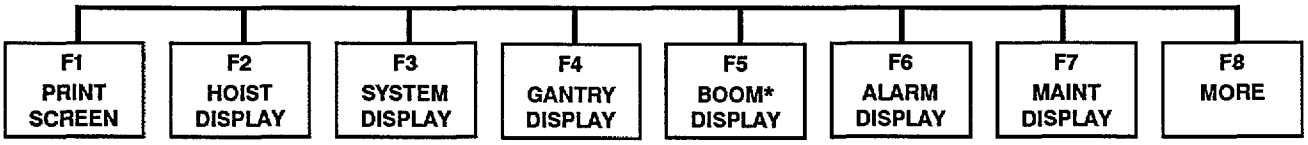
# SCREEN LEVELS (Continued)

## HOIST DISPLAY



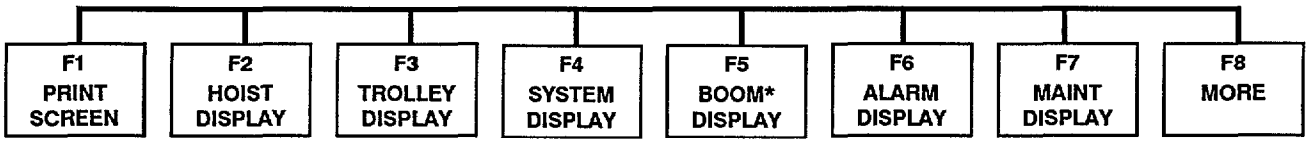

**Or "WHEEL TURNING"; see seperate screen example.*

## TROLLEY DISPLAY



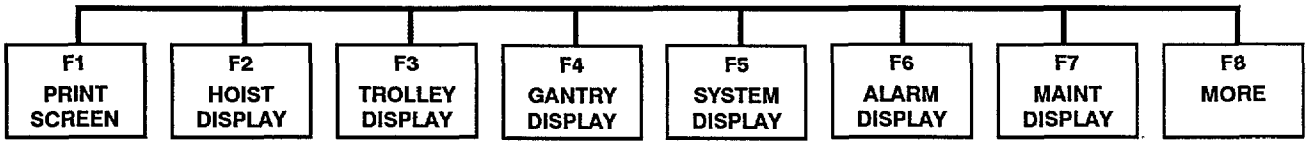

**Or "WHEEL TURNING"; see seperate screen example.*

## GANTRY DISPLAY

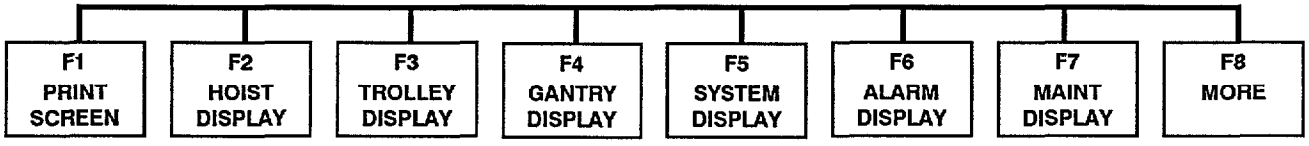



**Or "WHEEL TURNING"; see seperate screen example.*

## BOOM DISPLAY




## WHEEL TURNING DISPLAY

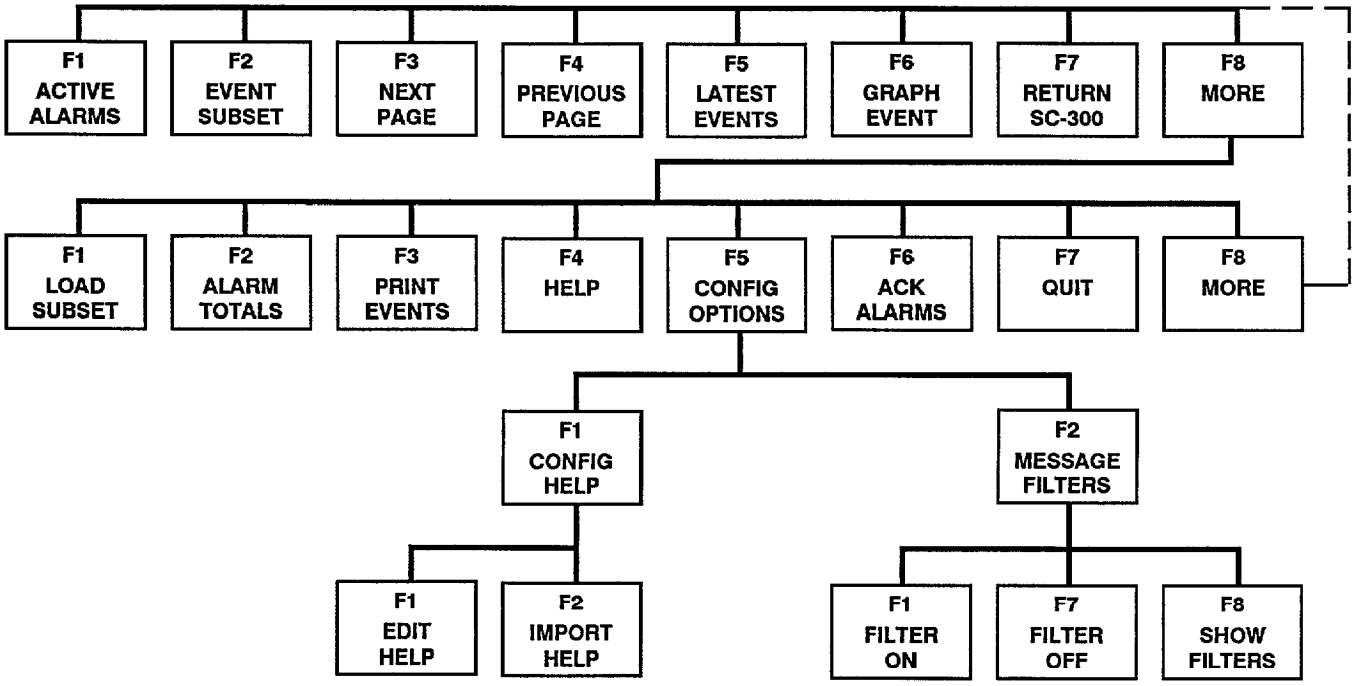


# SCREEN LEVELS (Continued)

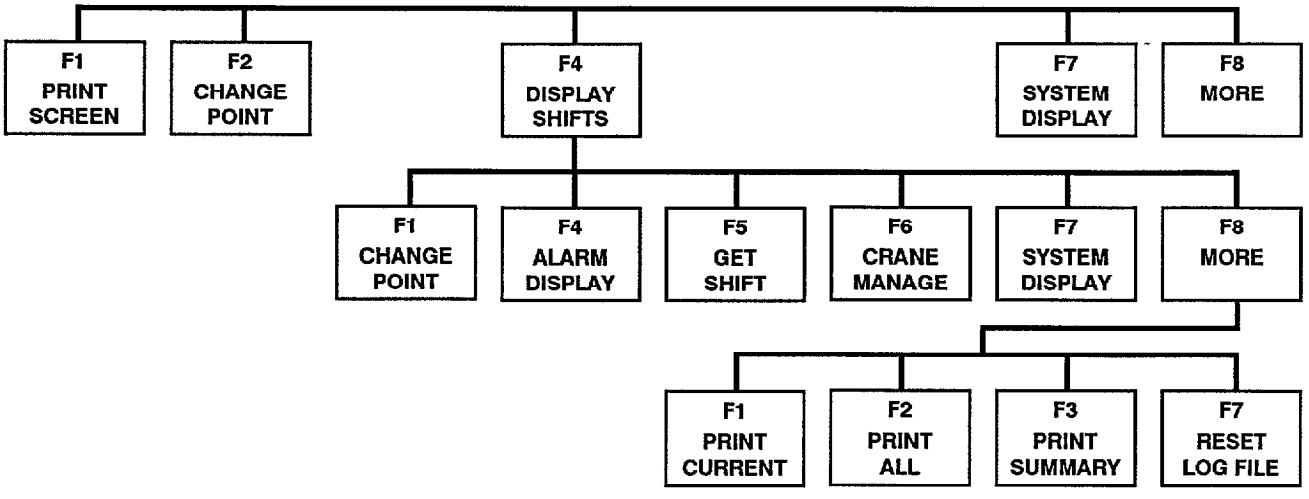
## MAINT DISPLAY



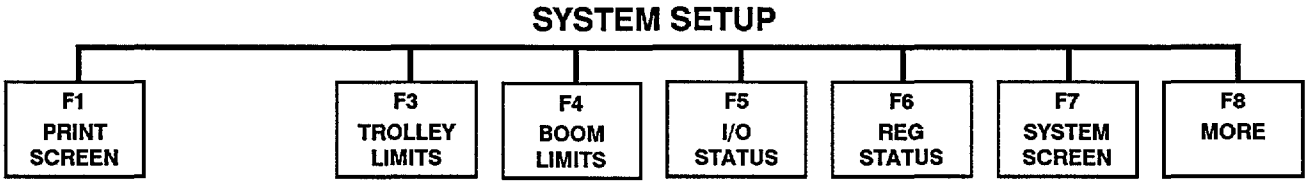

## ALARM DISPLAY

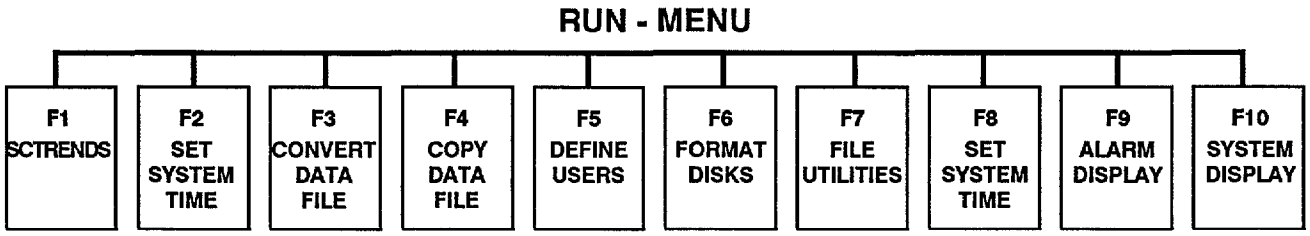



## CRANE MANAGEMENT



# SCREEN LEVELS (Continued)



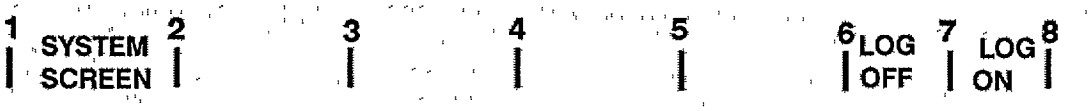

## *NOTES:*

# 4. STARTING UP THE ECMS

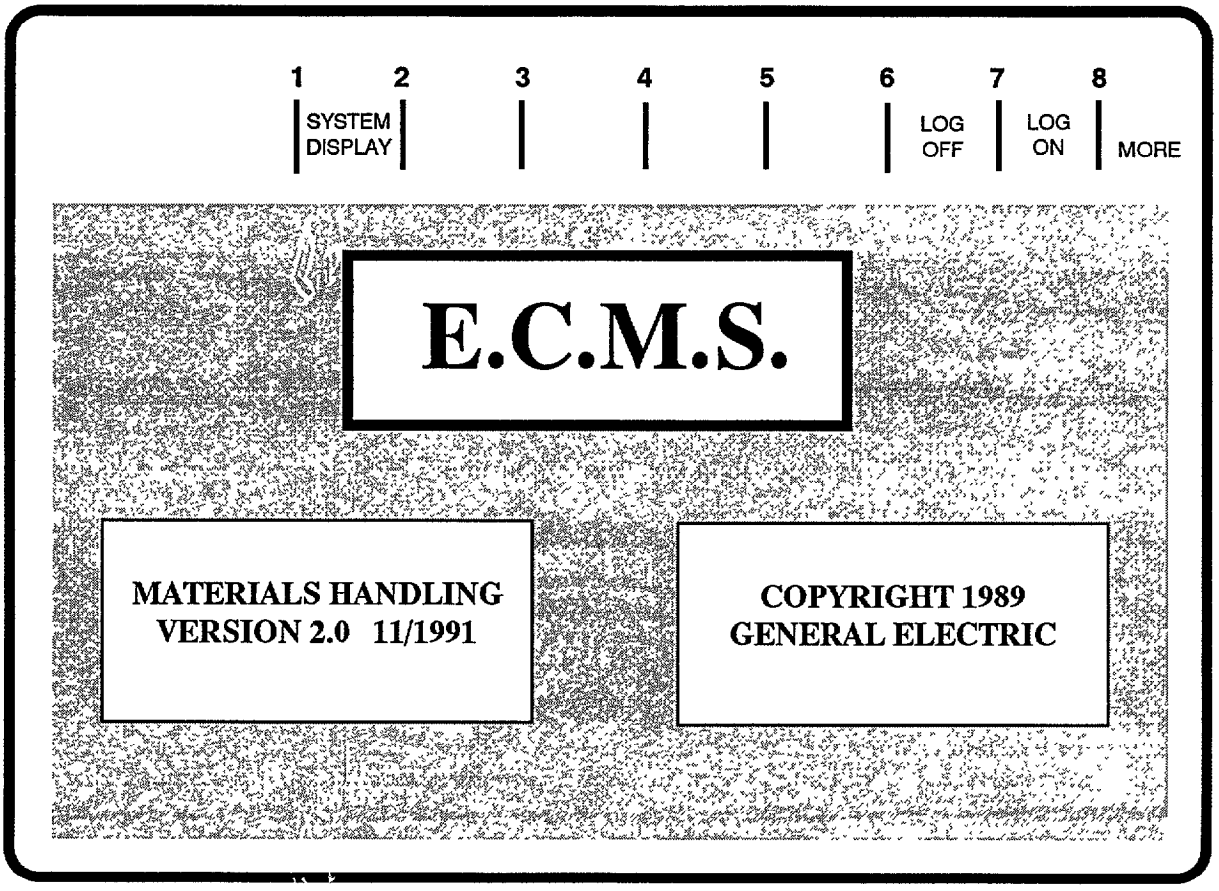
Once power is applied to the **ECMS** and the system is turned "ON", the computer will proceed through a self-test routine, then load the Operating System from the hard disk. At this point, a question will flash on the bottom of the screen as follows:

**LOAD CONCURRENT DOS [Y]:**

Unless interrupted by answering **"N"**, within a few seconds the computer will proceed to load the **ECMS** program into memory and bring up the **Introduction Screen**. This screen will remain on the monitor until the **ECMS** Software is fully loaded. After this, the **"TITLE"** Screen will appear. Function Keys are those keys listed **F1** through **F12** along the top of the keyboard. The **ECMS** normally only uses Function Keys **F1** through **F8**, and the action each key will perform is indicated along the top of the monitor's screen.




Pressing key **F1** will bring up the main **"SYSTEM"** Screen on the monitor. Pressing **F7** will allow a password user to **"LOG ON"** to the **ECMS**, and use those features not normally needed by regular maintenance staff. Pressing **F6** will **"LOG OFF"** or remove a password user from the system. The user may return to this screen from any **ECMS** Monitoring Screen by pressing Function Key **F10**.



# *NOTES:*

# 5. MONITORING

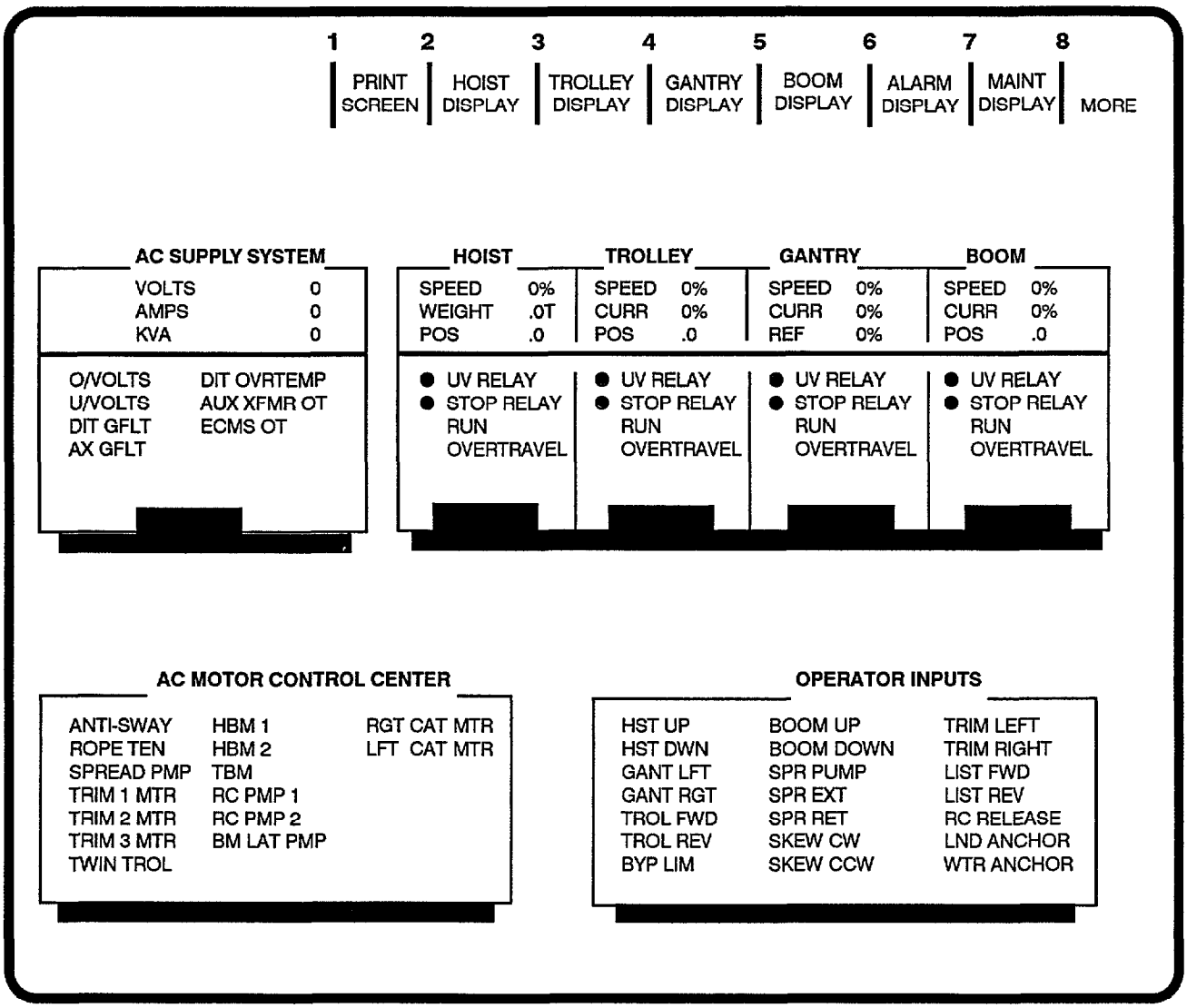
The Monitoring Function of the **ECMS** is performed by the **"SYSTEM"** Screen and the associated **"DRIVE"** Screens. Each of these screens is made up of **Background Data** (such as lines and text that does not change), **Foreground Data** (such as numbers and indicators), **Function Keys** along the top of the screen, and the **Computer Console** along the bottom of the screen. Generally, maintenance staff need not be concerned with the **computer console** line unless a computer warning message appears there.

## THE SYSTEM SCREEN

The **"SYSTEM"** Screen is divided into four main areas - AC SUPPLY, DC SYSTEM, AC MOTOR CONTROL, and OPERATOR inputs. Analog data is presented in the **blue** boxes in the AC SUPPLY and DC SYSTEM areas.

## COLOR CODING

The colors represented on the Monitoring Screens indicate the condition of a point. Those points that represent Alarm Conditions (eg. O/Volts) will be indicated either **green** for healthy, or **red** for Fault. Those points whose condition is merely ON or OFF, such as commands or motors, are indicated as **blue** for "ON" or "running", and **gray** (or blank) for "OFF".



## DRIVE SCREENS

Each of the **"DRIVE"** Screens (**HOIST, TROLLEY, GANTRY**, and **BOOM** - if provided) expands on the information previously available on the **"SYSTEM"** Screen. The layout of the areas on this screen is determined by the flow of control required to get that particular motion to operate. The top portion of the screen indicates the drive or motion name. The drive motion screens all appear the same but will be relative to the applicable drive motion.

The **STATUS** area indicates the drive motions variable feedback data. This typically includes percent speed, percent current, the crane number (for systems with multiple cranes), elapsed time, position, and percent speed reference.

The **STATUS** area indicates the drive motions discrete feedback data. This typically includes the MD Interlock, the Brake Release Interlock, the Standby Timer Status, and Encoder Status. When the particular status is ON, a **blue** status "bullet" will appear.

The **STATUS** area indicates the drive motions commands. This typically includes the motion signals and any other command applicable to that particular motion. When the particular status is ON, a **blue** status "bullet" will appear.

The **"UNDERVOLTAGE RELAY"** area indicates the status of the UV Relay. When the UV Relay is "picked up" or ON, the status "bullet" will be **green**. When the UV Relay is "dropped out" or OFF, the status "bullet" will be **red**. Whenever the UV Relay is OFF, there will be a condition which will have a **red** status "bullet" in the UV column. This condition indicates why the UV Relay may have "dropped out" or is OFF. Since each system has only one UV Relay, this portion of each Drive Motion Screen is identical.

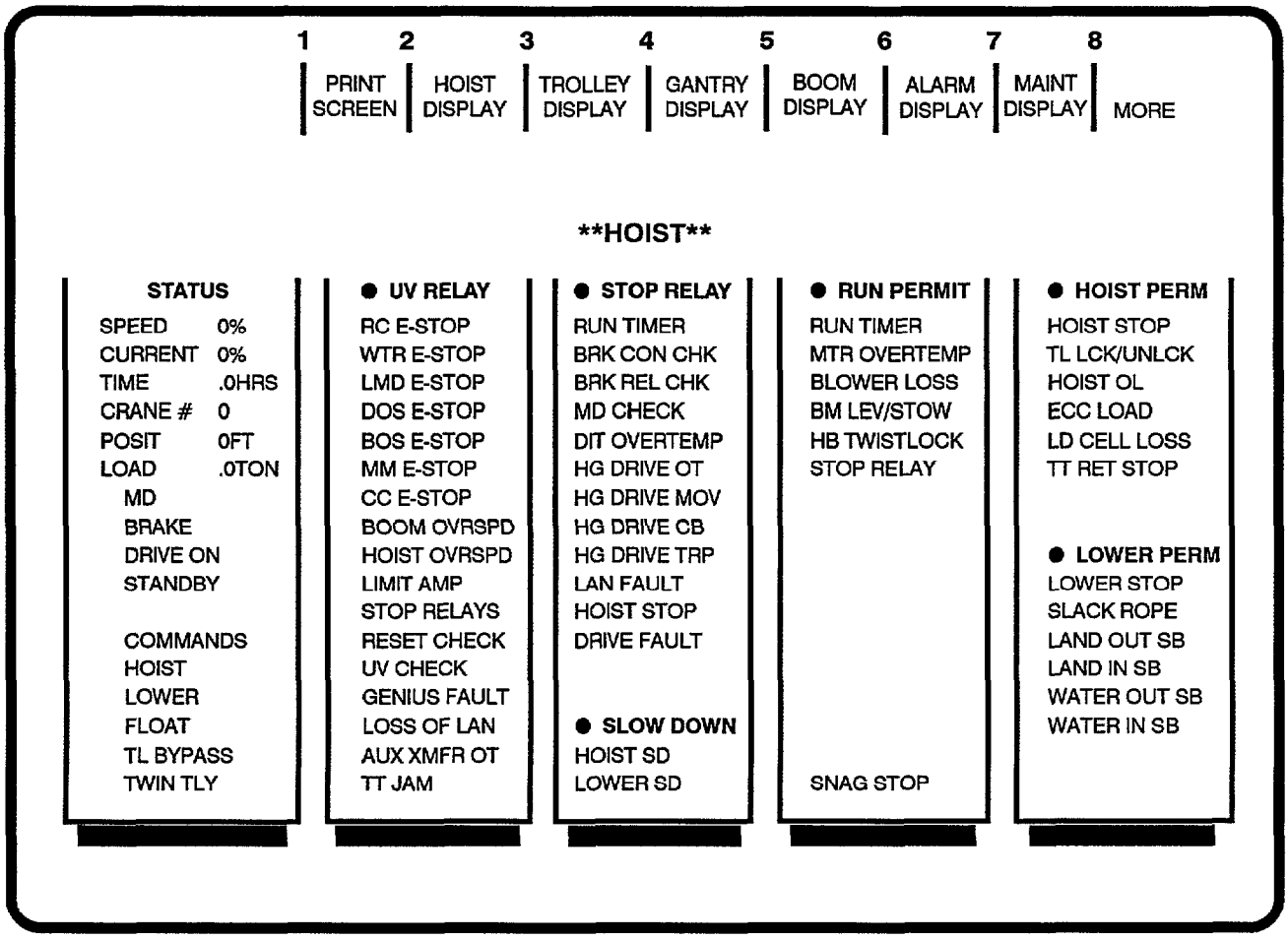
The **"STOP RELAY"** area indicates the status of the Drive Motions Stop Relay. When the Stop Relay is "picked up" or ON, the status "bullet" will be **green**. When the stop relay is "dropped out" or OFF, the status "bullet" will be **red**. Whenever the Stop Relay is OFF, there will be a condition that will have a **red** status "bullet" in the Stop Relay column. This condition indicates why the Stop Relay may have "dropped out" or is OFF.

The **"SLOWDOWN COMMAND"** area indicates the status of the Drive Motions Slowdown Command. When the Slowdown Command is "picked up" or ON, the status "bullet" will be **red**. When the Slowdown Command is "dropped out" or OFF, the status "bullet" will be **green**. When the Slowdown Command is ON, there will be a condition that will have a **red** status "bullet" in the Slowdown Command column.

The **"RUN PERMIT"** area indicates the status of the Drive Motions Run Permit Coil. When the Run Permit Coil is "picked up" or ON, the status "bullet" will be **green**. When the Run Permit Coil is "dropped out" or OFF, the status "bullet" will be **red**. Whenever the Run Permit Coil is OFF, there will be a condition that will have a **red** status "bullet" in the Run Permit column. This condition indicates why the Drive Motions Run Permit Coil may have "dropped out" or is OFF.

The **"DIRECTIONAL PERMIT"** area indicates the status of the Drive Motions Directional Permit Coil. When the Directional Permit Coil is "picked up" or ON, the status "bullet" will be **green**. When the Directional Permit Coil is "dropped out" or OFF, the status "bullet" will be **red**. When the Directional Permit Coil is OFF, there will be a condition that will have a **red** status "bullet" in the Directional Permit column. This condition indicates why the Drive Motions Directional Permit Coil may have "dropped out" or is OFF.

Pressing the **"PRINT SCREEN"** Function Key **F1** will create a printout of the present screen on the attached printer (if supplied).


1 PRINT SCREEN
2 HOIST DISPLAY
3 TROLLEY DISPLAY
4 GANTRY DISPLAY
5 BOOM DISPLAY
6 ALARM DISPLAY
7 MAINT DISPLAY
8 MORE
**HOIST**
STATUS
SPEED 0%
CURRENT 0%
TIME .0HRS
CRANE # 0
POSIT 0FT
LOAD .0TON
MD
BRAKE
DRIVE ON
STANDBY
COMMANDS
HOIST
LOWER
FLOAT
TL BYPASS
TWIN TLY
● UV RELAY
RC E-STOP
WTR E-STOP
LMD E-STOP
DOS E-STOP
BOS E-STOP
MM E-STOP
CC E-STOP
BOOM OVRSPD
HOIST OVRSPD
LIMIT AMP
STOP RELAYS
RESET CHECK
UV CHECK
GENIUS FAULT
LOSS OF LAN
AUX XMFR OT
TT JAM
● STOP RELAY
RUN TIMER
BRK CON CHK
BRK REL CHK
MD CHECK
DIT OVERTEMP
HG DRIVE OT
HG DRIVE MOV
HG DRIVE CB
HG DRIVE TRP
LAN FAULT
HOIST STOP
DRIVE FAULT
● SLOW DOWN
HOIST SD
LOWER SD
● RUN PERMIT
RUN TIMER
MTR OVERTEMP
BLOWER LOSS
BM LEV/STOW
HB TWISTLOCK
STOP RELAY
SNAG STOP
● HOIST PERM
HOIST STOP
TL LCK/UNLCK
HOIST OL
ECC LOAD
LD CELL LOSS
TT RET STOP
● LOWER PERM
LOWER STOP
SLACK ROPE
LAND OUT SB
LAND IN SB
WATER OUT SB
WATER IN SB

# *NOTES:*

# 6. DATA LOGGING

The **ECMS** assists in diagnostics and management, and is designed to capture data and retain it for review at a later date. The **ECMS** logs data in three ways:

An **ALARM** Log
An **EVENT** Log
**PEN RECORDINGS**

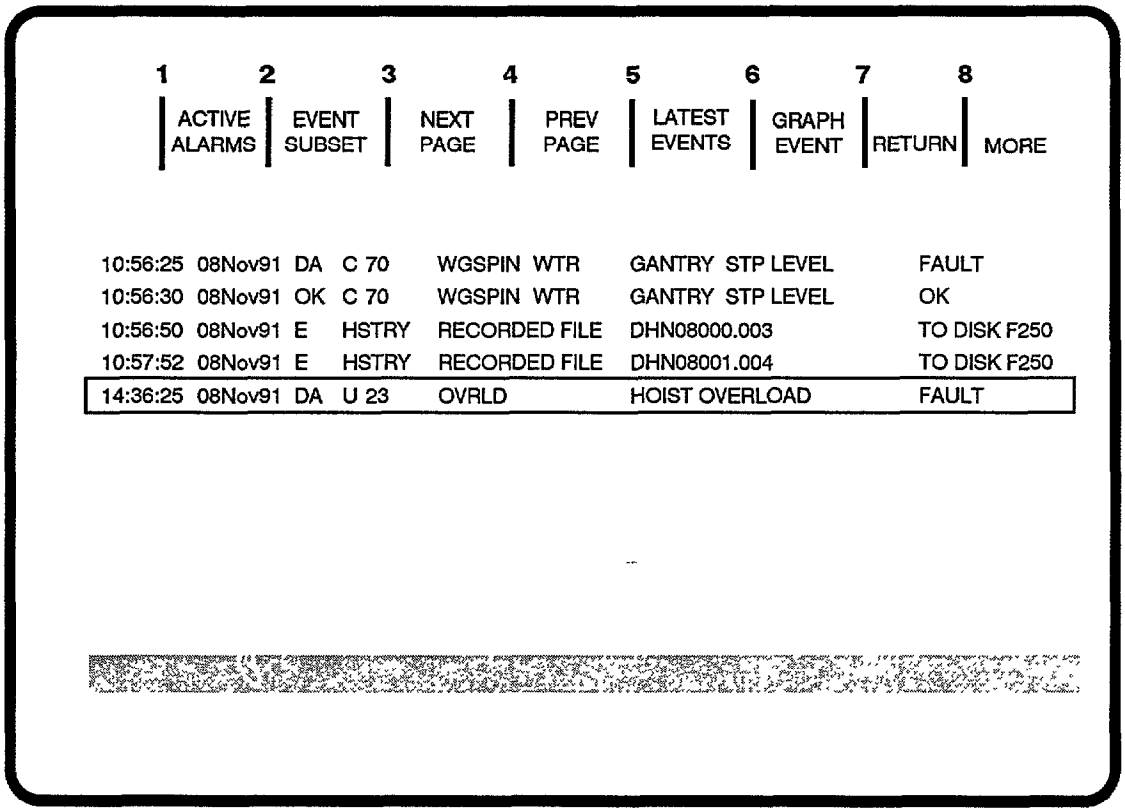
**.NOTE:** *The ECMS also has access to the drive's internal history logs.*

## EVENT/ALARM LOG:

The **ECMS** will keep a running log of the last 4000 programmed alarms and events that occurred on the crane. When the system was configured for this crane, some contacts changing state (eg. ON or OFF) were configured as **"EVENTS"** to be logged, some were configured as **"ALARMS"** to be logged, and some were for "Indication" on the Monitoring Screens only. Those contacts configured as **"EVENTS"** will appear in the **EVENT LOG**. Those contacts configured as **"ALARMS"** will appear in the **ALARM LOG** *and* the **EVENT LOG**.

This "Log" is a circular log file that makes an endless loop 4000 units long. As each **ALARM** and **EVENT** occurs, it is added to the list until four thousand have been added. At this point, the first entry is deleted to make room for the 4001 entry, and so it continues. The list contains the last 4000 **ALARMS** and **EVENTS** that have occurred on the crane. ALARMS and EVENTS occuring before that are lost.

The **ECMS** has the ability to detect Alarm Conditions in the system being monitored. In addition, many operator changes are detected as **EVENTS**. All of these **ALARMS** and **EVENTS** are available to the operator through the **ECMS Alarm Console**. **ALARMS** and **EVENTS** can occur for a variety of reasons in the system during normal operation. All of these **ALARMS** and **EVENTS** are translated into a message text that is sent to the **Alarm Console Program**. The **Alarm Console Program** provides a variety of user commands using Function Keys to display these text messages.



## ACTIVE ALARM DISPLAY

When the **"Event Console"** first starts up, it reads the stored **Alarm History** into memory and displays a message stating that initialization is in progress. When it has completed, the **Event Console** will be left showing the currently **ACTIVE ALARMS**. **ACTIVE ALARMS** are those messages that have either a priority of **URGENT** or **CAUTION**. When an **URGENT** or **CAUTION** Alarm occurs, it is always shown on the **ACTIVE ALARM DISPLAY**. When it returns to normal, it is removed from the **ACTIVE ALARM DISPLAY**.

The **"Event Console"** can retain up to 2000 **"High Priority"** Alarms for this display. When more than 2000 **"High Priority"** Alarms exist, the oldest alarms are removed from the list. The top of the Display Screen shows the available Functions to the operator. Each Function is selected by pressing the appropriate Function Key. The bottom of the Display shows the **Current Event Console Display** at the left side and the **Top Event Number** on the right side

The **"Top Alarm"** is a number representing a pointer into the list of 2000 **High Priority Alarms**. As you scroll through the list, the **Top Alarm Number** changes to indicate where you are in the list. Several keys are available to look at the **Active Alarms**. The **PGUP** and **PGDN** Keys cause the Display to page through a screenful at a time. The **CARRIAGE RETURN** Key scrolls up one line at a time. The **UP** and **DOWN CURSOR** Keys allow moving up and down one line at a time. The **HOME** Key brings you back to the **Top Alarm**.

**The function keys are shown below:**

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| ACTIVE ALARMS | EVENT SUBSET | NEXT PAGE | PREV PAGE | LATEST EVENTS | GRAPH EVENT | RETURN SYSTEM | MORE |

**Active Alarms**
This Function will change the Display to the **ACTIVE ALARM** Display.

**Event Subset**
This Function will display all **ALARMS** and **EVENTS** that fulfill the search conditions specified in the **"LOAD SUBSET"** System.

**Previous Page**
This Function will cause the screen to display the previous 22 **ACTIVE ALARMS**. This Function is inactive when the Display is already at the **Top Alarm Number 1**.

**Next Page**
This Function will cause the screen to display the next 22 ACTIVE ALARMS. This function is inactive when the display is already at the last alarm (#1000).

**Latest Events**
This Function brings the first display back up and sets the **TAB BAR** to the latest **ALARM/EVENT** on the list. This Function Key should always be used first when looking at the **EVENT Console**.

**Graph Event**
This Function is used to graph drive histories that have been stored on the hard disk by the **HISTORY** Program. A message indicating a **Drive History** was stored will appear on the **EVENT Console**. The operator should **TAB** to that **EVENT**, then use this Function Key. This will cause the **TOP VIRTUAL CONSOLE** to switch to **CONSOLE 2**. **CONSOLE 2** is reserved for running various **ECMS** Analysis Programs such as **SCTRENDS** or **STATDUMP**.

**Return to System**
This Function causes the **TOP VIRTUAL CONSOLE** to switch to **CONSOLE 1. CONSOLE 1** is reserved for running the main **ECMS Console Program.**

**More**
This Function Key will access the next level of functions available. They appear below.




**Load Subset**
This Function is used to interrogate the **EVENT History** for specific **ALARM/EVENT** occurences.

**Alarm Totals**
This Function will total all occurences of all **ALARMS** contained in the **History Files.**

**Print Events**
This Function will allow printing of EVENTS on the system printer as selected by the operator. All **ALARMS** and **EVENTS** will automatically be printed as they occur if the system printer is left on-line. This Function is to obtain additional printouts.

**Help**
This Function provides a **HELP SCREEN** for the **ALARM/EVENT** highlighted by the **TAB** Bar.

**Acknowledge Alarms**
This Function is used to reset Genius alarms inside **ECMS** and clear them from the **ACTIVE ALARM DISPLAY.** It will not reset the actual Genius block - this must be done with the **Hand-Held Monitor.**

## ALARM HISTORY DISPLAY

This Display is used to show **all** of the **System EVENTS and ALARMS** (including return to normal messages). The Console can display the 2000 most recent **ALARMS** and **EVENTS** and also records the last 2000 messages to a disk file (**EVENT.HIS**). When more than 2000 messages exist, the oldest messages are removed from memory. When more than 4000 messages have been recorded, the oldest are discarded.

The top of the **Display Screen** shows the available Functions to the operator. Each Function is selected by pressing the appropriate Function Key. The bottom of the Display shows the **Current ALARM Console Display** at the left side and the **Bottom ALARM Number** on the right side. The **Bottom ALARM** is a number representing a pointer into the list of 4000 **ALARMS** and **EVENTS.** As you scroll through the list, the **Bottom ALARM Number** changes to indicate where you are in the list.

Several keys are available to look at the **ALARM HISTORY.** The **PGUP** and **PGDN** Keys cause the Display to page through a screenful at a time. The **CARRIAGE RETURN** Key scrolls up one line at a time. The **UP** and **DOWN CURSOR** Keys allow moving up and down one line at a time. The **HOME** Key brings you back to the **Bottom ALARM.**

**The function keys are shown below:**



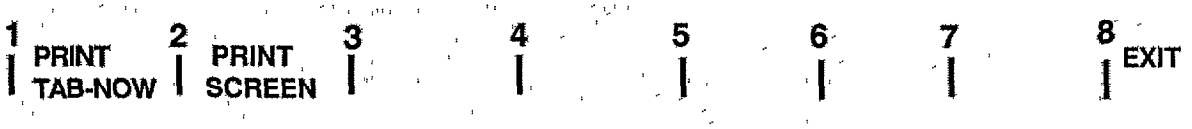
While most of these Functions have been previously described, several new ones are provided in **ALARM HISTORY Display** menu. These are described in the following paragraphs.

**Event History**
This Function changes the Display back to the **EVENT HISTORY Display.**

**Latest Alarms**
This Function sets the **Bottom ALARM** back to 200 so that the 22 most recent alarms are shown on the screen.

**Printing Alarms**
This Function, accessed from the **"MORE"** section of the above menu, causes a new set of Function Key actions to be displayed. The hard copy device chosen is the default printer for **VIRTUAL CONSOLE 3.** **ALARMS** that occur while printing a report are not recorded or displayed by the **ALARM Console.**

**This new menu is shown below:**




**Print TAB-NOW**
This Command will print the currently highlighted **ALARM/EVENT** line throught to the latest **ALARM/EVENT.**

**Print Screen**
This Command will print only those messages contained on the currently displayed screen.

## ALARM MESSAGE FORMAT(S)

This paragraph is provided to help understand the information that is contained in each of the various types of **ALARM** and **EVENT** messages. Each category of **ALARM** or **EVENT** in the system has a fixed format. The following Table identifies what type of message format is being used and how to decode the various fields in the message.

Each **ALARM** begins with the time and date that the **ALARM Message** text was generated. This is usually within 1 second of the actual **ALARM** detection. The next two to four characters define the general type of **ALARM** that is being shown. The possible codes are as follows:

| Type | Code | Description |
|---|---|---|
| 1 | **OK** | Return to normal message |
| 2 | **DA** | Digital Alarm |
| 3 | **DK** | Digital Acknowledge (reserved for future use) |
| 4 | **HI** | High Alarm |
| 5 | **HK** | High Alarm Acknowledge (reserved for future use) |
| 6 | **LO** | Low Alarm |
| 7 | **LK** | Low Alarm Acknowledge (reserved for future use) |
| 8 | **MDCH** | Drive Mode Change Alarm |
| 9 | **SPCH** | Setpoint change message |
| 10 | **GENI** | Genius Alarm |
| 11 | **##** | Drive fault or ECMS Program alarm |

**Type 11 ALARMS** use a two digit number in conjunction with the text string following it to indicate what drive or **ECMS** Program has generated the Fault message. **ALARM Types 1,8,9,** and **10** are considered **EVENTS** and are **not displayed** on the **ACTIVE ALARM Summmary Display.** (This is due to no return to normal condition for these **ALARMS** being logged.)

**The following paragraphs describe each alarm category:**

**Digital Alarms:** *15:20:38 03Feb88 DA U 272 COMFAIL COMMUNICATIONS STALLD*
The above example is used for **DIGITAL ALARMS. DIGITAL ALARMS** are used for indicating Faults when one state indicates Normal Status, and the other is a Fault Condition.

The **ALARM Priority** is indicated by the **"U"** character*. In this example, the priority is **URGENT**. The Number of the **ALARM** is **272**. This Number can be used to disable the **ALARM** from the **ECMS Console**, if desired. The **COMFAIL Text** is a **1** to **9** character "Nickname" (**Tagname**) that is unique for every alarm point in the system. It is followed by an english description of the **ALARM Point** and its new state. In this case, the state is **STALLD**. When the **ALARM** returns to normal, the state will indicate **HEALTHY**.

**Analog Alarms:** *5:27:00 03Feb88 LO C E1B1TFB EXTRUDER 1 TEMP FDBK 100.00*
The above example is used for **ANALOG ALARMS. ANALOG ALARMS** are similar to **DIGITAL ALARMS** except that they provide the current value of the point when the **ALARM Text Message** was created. In addition, the engineering units for the point are shown immediately following the current value.

**Genius Alarms:**
**GENIUS ALARMS** are always indicated by the word **GENIUS** in the Description Field. **GENIUS ALARMS** can indicate Faults for Bus Controller(s), Blocks, and Points. **GENIUS ALARMS** must be supported in the Series Six Programmable Controller Ladder Logic Program in order for these messages to be generated.

**Drive Alarms:** *15:27:00 03Feb88 4 U HOIST LOSS OF AN AC PHASE F22*
The above example is used for **DRIVE ALARMS.** The **Drive Drop Number** on the **LAN** is indicated by the number **4.** The description of the **ALARM** includes an english description of the drive along with an expanded Fault Description. The **Fault Number** is shown at the end of the message. (See appropriate Drive Instruction Book for Fault Numbers).

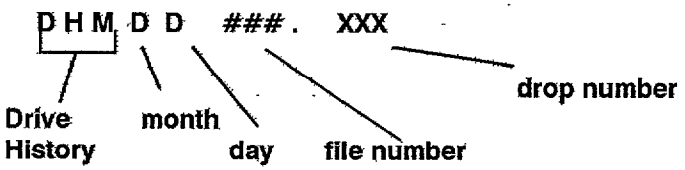
**Setpoint Change Messages:**
*15:25:06 03Feb88 SPCH HMREC HOIST MOTOR REC HOURS 0000 TO 1200 HRS*
*15:26:01 03Feb88 SPCH TMREC TROLLEY MOTOR REC HOURS 0000 TO 950 HRS*
The above examples are for **SETPOINT CHANGE MESSAGES.** Whenever a point is changed by the operator using the **CHANGE POINT** Function Key, an english message **ALARM** is generated. This message records the **SETPOINT CHANGE** showing the "Nickname" (**Tagname**), its description, the old value, and the new value with its engineering units (if applicable).

**ECMS Program Alarms:**
*15:23:49 03Feb88 72 E SC-SETUP Execution from DISK complete F93*
*15:27:00 03Feb88 75 E RECORDER Pen channel started and sampling F203*
The above examples are for **ECMS PROGRAM ALARMS.** Several of the programs in the **ECMS System** generate **ALARM** or **EVENT** messages to indicate System Conditions. For example, when a new pen is connected in the Recorder System, an **EVENT Message** is generated indicating that a Pen Channel has started. When recipe operation is complete, an **EVENT Message** is generated to indicate the request has been processed.

**E is EVENT, C is CAUTION, and A is ATTENTION.*

## DRIVE HISTORY MONITORING

The program in ECMS that collects the "Drive History" is called **HISTORY** and operates in the background while **ECMS** is working. **HISTORY** will continually scan all drives to see if an **Internal Drive History** has been locked (stopped recording). If for any reason the **Drive History** has been locked, **HISTORY** will record the drive history into a raw data file with a file naming convention as follows:




Once the file is recorded to disk, **HISTORY** will attempt to unlock the **Drive History**. If this fails, **HISTORY** will not record another file from this drive until unlocking is successful. **HISTORY** will also log an **EVENT** to the **ALARM Log** when a **Drive History** is read.

Each drive has the ability to store a **"Mini-History"** within its own memory. This history can be configured in one of two forms:

* **50 Readings Each of Five Variables** (Factory default is 50 readings with five variables)
* **100 Readings Each of Two Variables**

Exactly which of the many possible variables within the drive are stored in its **"Mini-History"** is determined by the software parameters in the drive. The following parameters set variable sampled and time base for sampling:

The list of **RAM Locations** and their associated variable are as follows:

| VARIABLE | RAM | VARIABLE | RAM |
|---|---|---|---|
| ACOM | **494** | LNSPD | **1096** |
| ANAJR | **465** | LSCMP | **1106** |
| ASP0 | **379** | LTA | **413** |
| ASP2 | **461** | Net Speed Ref. | **63** |
| ASR | **486** | OR Bias | **1073** |
| Blk Comp | **1146** | Position | **453** |
| CEMF | **105** | Posn Loop Error | **471** |
| Curr Feedback | **55** | Posn Loop Integ | **480** |
| Curr Loop Out | **61** | Posn Loop Output | **482** |
| CLA | **414** | PSREF | **469** |
| CT Phase 1 | **428** | Pulse Ref Freq | **83** |
| CT Phase 3 | **429** | Real Time Clock | **381** |
| Current Offset | **28** | Ref Total | **1075** |
| Current Ref | **443** | Speed Feedback | **71** |
| Digital Tach. | **77** | Spfb with Lead | **73** |
| DVM | **496** | SJ1 | **459** |
| Encr. PI Posn | **91** | SJ2 | **457** |
| Fault Code | **23** | Speed Droop | **1038** |
| Field Flux | **1114** | Speed Loop Error | **101** |
| Field Current | **463** | Spd Loop Integr | **51** |
| Filtered CFB | **448** | Spd Loop Output | **53** |

| VARIABLE | RAM | VARIABLE | RAM |
|---|---|---|---|
| Filtered SFB | 484 | Spd Loop Propor | 48 |
| Filtered VFB | 445 | Tach Foll. Error | 89 |
| Firing V-Cmd | 30 | Tapered Curr Lim | 18 |
| FRA | 467 | Timed Speed Ref | 65 |
| FUP Diag. Var. | 93 | TIR | 1050 |
| Horsepower | 492 | TOC Timer | 501 |
| I ABS | 57 | Torque | 455 |
| IREF | 1100 | TUP Diag Var | 87 |
| JCOMP | 1102 | Volt Feedback | 103 |
| Jog Reference | 69 | VLOPO | 475 |

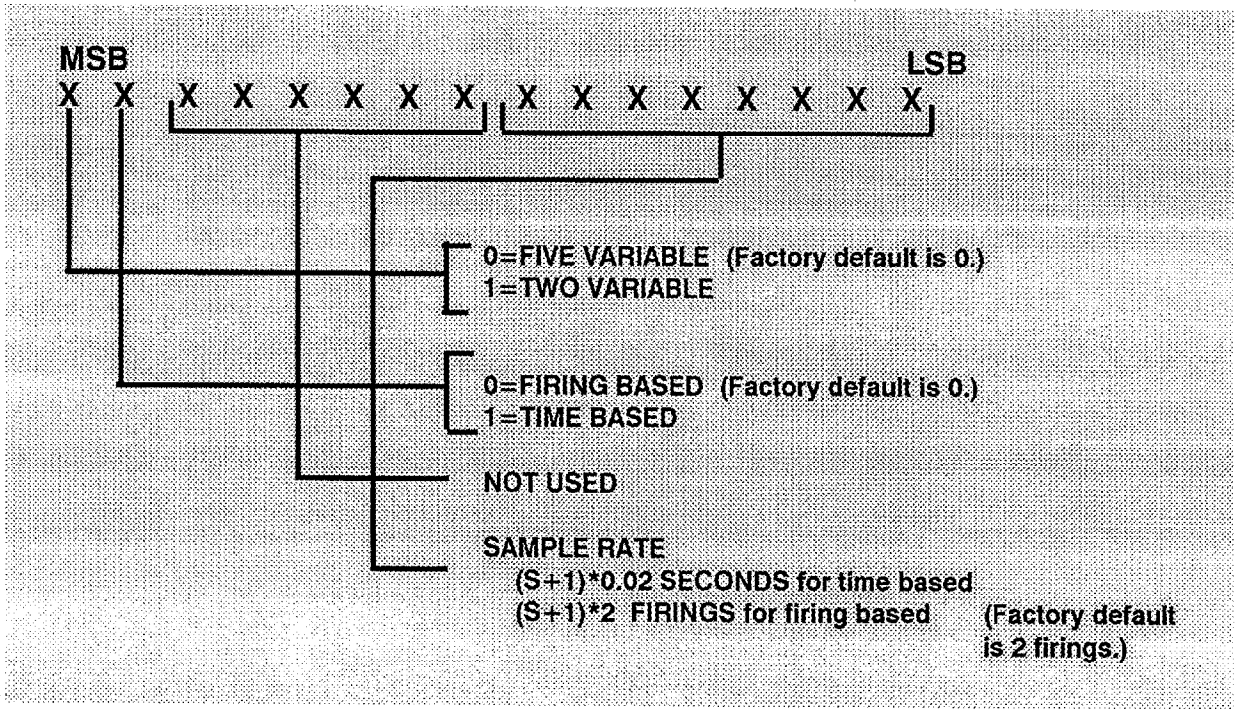
In addition to the previous **"Continuous" Variables**, certain **"Bit" Varaibles** are available to the **HISTORY** recording facility:

| VARIABLE | RAM | VARIABLE | RAM |
|---|---|---|---|
| Adv Rate Lim | **241** | Precondition | **152** |
| Auto | **146** | Reference Enable | **148** |
| CEMF Lim Flg | **141** | RSTOP satisfied | **147** |
| Curr limit Flag | **140** | Run 0 | **190** |
| Curr Rate Limit | **240** | Run 1 | **191** |
| Disc. Curr Flag | **157** | Shunt Trip | **130** |
| Firing Enabled | **139** | Slow | **194** |
| I Follow Req. | **162** | Torque Proved | **255** |
| I Follow Active | **172** | Trip Fault | **137** |
| No Fault | **136** | Up to Speed | **155** |
| Posn Loop Lim | **251** | Zero Curr Flag | **156** |
| Posn Loop Run | **250** | | |

## EEROM PARAMETERS REQUIRED FOR HISTORY

**EE378 - HISTD**

This **EEROM Location** allows the user to select the type of **DRIVE HISTORY** desired (Two Variable or Five Variable). A **Two Variable History** will record **102** samples of any two variables, a **Five Variable History** will record **51** samples with two of the five variables being selectable (the other three variables being Current Feedback, Voltage Feedback, and RUP Status). The user can select the histories to be **SCR Firing** or **Time Based. SCR Firing** sample times are multiples of every other firing. **Time Based** samples are multiples of 20 msec. A selection of 0 gives minimum sample time; a selection of 4 will give five times the sample period.




**EE379 HISPT**

This **EEROM Location** carries the **RAM Addresses** for the **two** selectable variables (either **1** and **2** for Two Variable History *or* **4** and **5** for Five Variable History). **Least significant** first eight bits of **EE379** is the first address; **most significant** eight bits is the second address.

**EE251 UAPJP**

This **EEROM Location** carries the **HISTORY LOCK** source. Histories may be locked manually or by selecting a locking trigger through this address. The selection can include:

* Any time the drive has a trip fault. (Factory Default)
* Any time SLD4 picks-up.
* Any time the drive has any type of fault.

Bits **8** and **9** of this **EEROM Location** are to be set as follows: (LSB = 0, MSB = 15)

| 8 | 9 | TRIGGER TYPE |
|---|---|---|
| 0 | 0 | Trip Fault Trigger (Factory Default) |
| 1 | 0 | SLD4 Trigger |
| 0 | 1 | Trip or Annunciated Fault Trigger |
| 1 | 1 | SLD4 Trigger |

**NOTE:** *To use **SLD4**, the user must set-up the **Signal Level Detector Parameters**. This set-up is detailed in the applicable Drive Instruction Book.*

**EE009 SLDJP**
This **EEROM Location** cannot be changed while the drive is operating. Bits **6** and **7** must be set to the table below for correct operation of **SLD4**. Note that setting these bits (as with all others) involves adding or subtracting a binary number from that already existing in this location.

| 6 | 7 | SLD ACTION |
|---|---|---|
| 0 | 0 | PU if EE118>EE119 |
| 1 | 0 | PU if \|EE118\|>EE119 |
| 0 | 1 | DO if EE118>EE119 |
| 1 | 1 | DO if \|EE118\|>EE119 |

**EE118 SLD4I**
The user must enter the monitored variable's **RAM Address** into this **EEROM Location**. (See previous **RAM Address** Tables).

**EE119 SLD4L**
The user enters the level that the signal must exceed to pick-up or drop-out **SLD4**.

**EE089 SLDTD**
The user may specify a **Time Delay** for **SLD4** if required. Bits **4** and **10** are set (by adding or subtracting a binary value) in the number in this location as follows:

| 4 | 10 | SLD ACTION |
|---|---|---|
| 1 | 0 | Enables Drop-Out Time Delay |
| 0 | 1 | Enables Pick-Up Time Delay |

**EE120 SLD4H**
This **EEROM Location** will contain the **Time Delay** to be used if enabled. The **least significant** eight bits contain the **Pick-Up Time Delay** in 10ths of a second (i.e., maximum of 25.5 sec possible). The **most significant** eight bits contain the **Drop-Out Time Delay** in 10ths of a second (again maximum of 25.5 sec possible).

## VIEWING HISTORICAL DATA (For technical information only)

The **UNLOAD Program** is a companion of the **HISTORY Program** and operates in background mode, transparent to the user. The sole purpose of this program is to convert the raw history files into a human readable form.

### SCTREND GRAPH FORMAT

**HISTORY** is unloaded to view in **ECMS. UNLOAD** will strip the **HISTORY File** of a selected variable and convert it into an **SCTREND Graph File**. Once these Graph Files have been created, the user then accesses the data from the message produced by **UNLOAD** on the **EVENT Console. SCTREND** automatically draws the data from the raw history file(s) onto the **Graph Window**. Unloading in this method normalizes variables to their Per Unit values rather than raw counts and inverts references and feedbacks so they are in the same polarity for better comparisons if the variables are known by the **UNLOAD Program.**

### UNLOAD - Further technical information

**Time Based HISTORY Variables** may be unloaded from different raw files with different lock times. The **UNLOAD Program** will automatically space the graphs correctly in time and zero suppress (buck-out) the variables for maximum resolution. **Firing-Based Samples** may only be unloaded from a single raw file for viewing since these variables can only be related to themselves in time and not to other histories loaded at different times.

The unloading of **HISTORY Files** for **SCTREND** will probably be the most used option of the **UNLOAD Program.** The UNLOAD for the printer or for **LOTUS** is fairly simple and only requires that the user know the name of the raw history file. For the **LOTUS** unloading, the file created by the **UNLOAD Program** that may be then used for analysis with **LOTUS 123** has the same name as the raw history file used except the first two characters (**DH**) are replaced with **LT**.

## PEN RECORDINGS

**Historical Logging Set-up**, or **Chart Recorder**, (accessed from the secondary set of Function Keys available on the **SYSTEM Screen** or **DRIVE Screens**) will store data in files that can later be used with the **Trendgraph Function**. The recorder allows for up to **64 "Pens"** to be assigned.

To assign a **"Pen"**, press the **TAB** Key **until** the Pen required has an arrow (=>) beside it on the screen. Then press the **ENABLE PEN** Function Key, and enter the **Tagname** of the variable you wish to log. (Valid tagnames can be found previously by pressing the **LIST TAGNAME** Function Key - Refer to Table on the following page for an explanation of Tagnames.)

Next, enter the **Sampling Period** for the Pen. The Sampling Period is the time between each reading of the variable (eg. a value of 1 makes the **ECMS** sample that variable every 1 second). A seperate **"PEN FILE"** will be created for each pen asigned. Each **PEN FILE** is a "rotating" store (or endless loop) **3600** readings long. Therefore, a one second sample period will eventually store **3600** seconds of data for that variable. The Sampling Period can range from **1 to 999 seconds,** so the total store length for each Pen can range from **1 hour to 999 hours.** Logging will continue until all Pens are disabled (or the logging is inhibited on the **ECMS STATUS SCREEN**). Pens are disabled by **"Tabbing"** to the Pen, then pressing the **DIS PEN** Function Key.

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| ENA/DIS PEN | SAVE SETUP | RECALL SETUP | PREV SCREEN | LIST TAGNAME | RETURN | GRAPH CONSLE | MORE |

| PEN NAME | PER (S) | PEN NAME | PER (S) | PEN NAME | PER (S) | PEN NAME | PER (S) |
|---|---|---|---|---|---|---|---|
| 1 HSFB | 1 | 17 | 1 | 33 | 1 | 49 | 1 |
| 2 TSFB | 1 | 18 | 1 | 34 | 1 | 50 | 1 |
| 3 HPOSN | 1 | 19 | 1 | 35 | 1 | 51 | 1 |
| 4 TPOSN | 1 | 20 | 1 | 36 | 1 | 52 | 1 |
| 5 | 1 | 21 | 1 | 37 | 1 | 53 | 1 |
| 6 | 1 | 22 | 1 | 38 | 1 | 54 | 1 |
| 7 | 1 | 23 | 1 | 39 | 1 | 55 | 1 |
| 8 | 1 | 24 | 1 | 40 | 1 | 56 | 1 |
| 9 | 1 | 25 | 1 | 41 | 1 | 57 | 1 |
| 10 | 1 | 26 | 1 | 42 | 1 | 58 | 1 |
| 11 | 1 | 27 | 1 | 43 | 1 | 59 | 1 |
| 12 | 1 | 28 | 1 | 44 | 1 | 60 | 1 |
| 13 | 1 | 29 | 1 | 45 | 1 | 61 | 1 |
| 14 | 1 | 30 | 1 | 46 | 1 | 62 | 1 |
| 15 | 1 | 31 | 1 | 47 | 1 | 63 | 1 |
| 16 | 1 | 32 | 1 | 48 | 1 | 64 | 1 |

# TAGNAME LISTING

The following list of Tagnames are all abbreviations of variables that are used in the system.

| Tagname | Description |
|---|---|
| **DRV1FC** | Drive 1 Fault Code (typically hoist/gantry drive). |
| **DRV2FC** | Drive 2 Fault Code (typically trolley/boom drive). |
| **WPPM** | Current month in system. |
| **WPDD** | Current day in system. |
| **WPYY** | Current year in system. |
| **OPHRS** | Current shift operating hours. |
| **DNHRS** | Current shift down-time hours. |
| **IDLHRS** | Current shift idle-time hours. |
| **TOLTIME** | Total time currently into this shift. |
| **OPSHIFT** | Current operating shift. |
| **LOADA** | Current number of smallest size containers loaded this shift. |
| **LOADB** | Current number of next size containers loaded this shift. |
| **LOADC** | Current number of next size containers loaded this shift. |
| **LOADD** | Current number of next size containers loaded this shift. |
| **LOADE** | Current number of next size containers loaded this shift. |
| **TOTLD** | Current number of containers loaded this shift. |
| **UNLDA** | Current number of smallest size containers unloaded this shift. |
| **UNLDB** | Current number of next size containers unloaded this shift. |
| **UNLDC** | Current number of next size containers unloaded this shift. |
| **UNLDD** | Current number of next size containers unloaded this shift. |
| **UNLDE** | Current number of next size containers unloaded this shift. |
| **TOTUL** | Total number of containers unloaded this shift. |
| **NETCONT** | Net container rate per hour (TOTLD + TOTUL/OPHRS). |
| **GROSCON** | Gross container rate per shift (TOTLD + TOTUL/Length of Shift). |
| **AVCYC** | Average cycle time to complete a load or unload operation. |
| **SAVWGT** | Summary of all shifts requested average weight containers. |
| **SPKWGT** | Summary of all shifts requested peak weight containers. |
| **SNETCO** | Summary of all shifts requested containers loaded and unloaded. |
| **ISDAY** | Individual shift requested current day. |
| **ISMON** | Individual shift requested current month. |
| **ISYEAR** | Individual shift requested current year. |
| **ISSHFT** | Individual shift requested current shift. |
| **ISLDA** | Individual shift requested smallest size containers loaded. |
| **ISLDB** | Individual shift requested next size containers loaded. |
| **ISLDC** | Individual shift requested next size containers loaded. |
| **ISLDD** | Individual shift requested next size containers loaded. |
| **ISLDE** | Individual shift requested next size containers loaded. |
| **ISOPERT** | Individual shift requested operating time. |
| **ISDOWNT** | Individual shift requested down-time. |
| **ISIDLET** | Individual shift requested idle-time. |
| **ISTOTLT** | Individual shift requested total time. |
| **ISNETCO** | Individual shift requested net containers loaded. |

# TAGNAME LISTING (continued)

| Tagname | Description |
|---|---|
| **ISAVWGT** | Individual shift requested average weight of all containers. |
| **ISPKWGT** | Individual shift requested peak weight of all containers. |
| **SSYY** | Start of first shifts year for shift reports. |
| **SSMM** | Start of first shifts month for shift reports. |
| **SSDD** | Start of first shifts day for shift reports. |
| **32STAT1** | Screen 32 status 1 (hoist) percent rated speed. |
| **32STAT2** | Screen 32 status 2 (hoist) percent rated current. |
| **32STAT3** | Screen 32 status 3 (hoist) crane number. |
| **32STAT4** | Screen 32 status 4 (hoist) elapsed time. |
| **32STAT5** | Screen 32 status 5 (hoist) position. |
| **32STAT6** | Screen 32 status 6 (hoist) percent speed reference. **NOTE:** *When hoist position is not available, percent speed reference will be 32STAT5.* |
| **34STAT1** | Screen 34 status 1 (trolley) percent rated speed. |
| **34STAT2** | Screen 34 status 2 (trolley) percent rated current. |
| **34STAT3** | Screen 34 status 3 (trolley) crane number. |
| **34STAT4** | Screen 34 status 4 (trolley) elapsed time. |
| **34STAT5** | Screen 34 status 5 (trolley) position. |
| **34STAT6** | Screen 34 status 6 (trolley) percent speed reference. **NOTE:** *When trolley position is not available, 34STAT5 may be left open.* |
| **36STAT1** | Screen 36 status 1 (gantry) percent rated speed. |
| **36STAT2** | Screen 36 status 2 (gantry) percent rated current. |
| **36STAT3** | Screen 36 status 3 (gantry) crane number. |
| **36STAT4** | Screen 36 status 4 (gantry) elapsed time. |
| **36STAT5** | Screen 36 status 5 (gantry) position. |
| **36STAT6** | Screen 36 status 6 (gantry) percent speed reference. **NOTE:** *When gantry position is not available, 36STAT5 may be left open.* |
| **38STAT1** | Screen 38 status 1 (boom) percent rated speed. |
| **38STAT2** | Screen 38 status 2 (boom) percent rated current. |
| **38STAT3** | Screen 38 status 3 (boom) crane number. |
| **38STAT4** | Screen 38 status 4 (boom) elapsed time. |
| **38STAT5** | Screen 38 status 5 (boom) position. |
| **38STAT6** | Screen 38 status 6 (boom) percent speed reference. **NOTE:** *When boom position is not available, 38STAT5 may be left open.* |

# 7. DIAGNOSTICS

Refer to **ALARM and EVENT LOG** Paragraph of **DATA LOGGING** for a complete description of the operation of these functions. When diagnosing a Fault in the **Crane System**, use the following procedure as a guide:

**STEP 1**

Examine the **ALARM and EVENT LOG** to determine the actual Fault or Faults indicated by the System. Depending on the source of the Fault and the part of the System reporting the Fault, these messages are likely to be very accurate. Some Faults may cause a chain of other **"Pseudofaults"** (apparent faults that don't really exist). Be aware of this possibility.

**STEP 2**

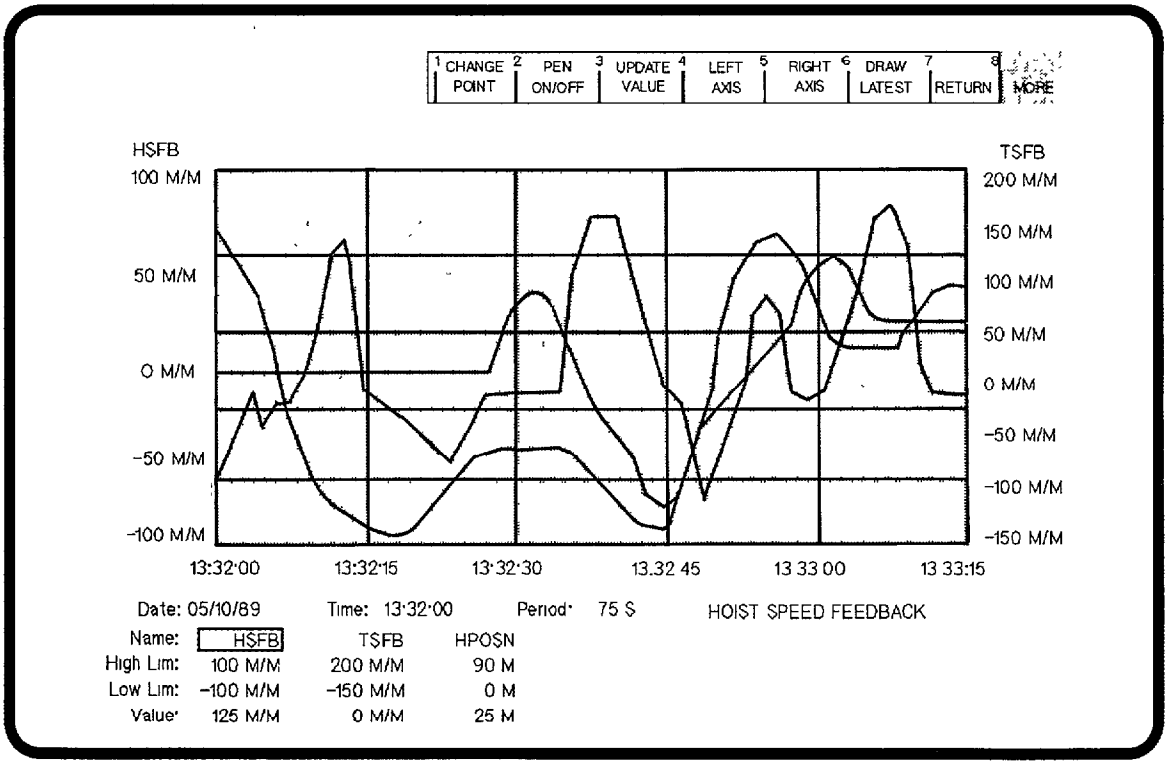
Sometimes an actual problem may not appear as a Fault. An example could be - No Output From The Masterswitch. Next examine the **MONITORING SCREENS** to check the sequence of operation of the particular section.

**STEP 3**

Certain Faults may be being caused by a sequence of events - examine the **EVENT HISTORY** to see if a Fault always occurs after certain **EVENTS** take place.

## TRENDGRAPHS

Sometimes, Faults are intermittant and seem to occur for no apparent reason. It may be helpful to set-up the pen **Recorder Function** to monitor a number of suspected variables or **"Unlogged" EVENTS** (**EVENTS** that are not logged in the **EVENT HISTORY**). Once the Recorder has been activated (refer to the **DATA LOGGING Section** of this manual), **PEN FILES** are created and continuously updated on the hard disk. These **PEN FILES** are used by the **SCTREND Function** to create graphic plots of the data.



## HOW TO USE THE SCTREND FUNCTION

The **SCTREND Function** is part of the **ANALYSIS Section** of the **ECMS**. Switch the **ECMS** to the **PEN RECORDER Section** and then press the **GRAPH** Function Key. Here, the user may be presented with a menu of Functions, the first being **"RUN SCTREND" (F1)**, or the Graph Program may already be running and available. It may also be useful to create and load a default graph using the **SAVE** and **RECALL** Functions of **SCTREND**.

To create a new graph, the user must first have stored some data in **PEN FILES** by using the **Historical Logging Set-up Screen**. **PEN FILES** are labeled **PEN.01, PEN.02**, etc. by the Recording System. Press the **TAB** Key until the **blue** cursor box appears next to **"Name:"**; then press the **CHANGE POINT** Function Key. Enter the name of the **PEN FILE** (eg. **PEN.01**); then press **ENTER**. If the file exists, the data will be plotted by **SCTREND** on the Screen. Further files can be retrieved by tabbing to the next **"Name:"** Field and repeating the proceedure.

To modify default graph settings to reflect the information actually required, use the **TAB** Key to move the blue cursor box over the data to be modified. Once the cursor is over the field to be changed, press the **CHANGE POINT** Function Key **F1**. The Trendgraph Program will prompt the user for the new data. If the cursor is over the **TIME Field**, the user will be prompted to enter a new graph time range. This should be entered in seconds, minutes, or hours as the following examples show:

**Example:**

**60mIn** = 60 minutes
**1hour** = 1 hour
**25sec** = 25 seconds
**235m** = 235 minutes

Only the first four digits are read; therefore, the time must be three digits or less (allowing one digit for **S** [seconds], **M** [minutes], and **H** [hours] at the end). The time period displayed on the graph will range from the date and time entered back in time by the time period entered.

### Changing The Y Axis High And Low Limits For A Pen

Tab to the desired Pen limits and depress **CHANGE POINT**. The new high and low limits will then be requested.

### Changing The Graph Start Date

The **Starting Date** is changed by tabbing to the **Date Field** and depressing the **CHANGE POINT** Key. At this time the desired date will be requested. The date may either be entered in **MM/DD/YY** format or as an incremental date such as **"-1D"** meaning 1 day older. The valid time and date suffixes are: **D:** day **H:** hour **M:** minute **S:** second. If a blank date is entered, the date and time will be set to the most current time in any of the currently displayed **PEN FILES**.

### Changing The Graph Start Time

The **Starting Time** is changed by tabbing to the **Time Field** and depressing the **CHANGE POINT** Key. At this time, the desired time will be requested. The time may either be entered in **HH:MM:SS** format or as an incremental time such as **"1H"** meaning 1 hour later. If a blank time is entered, the date and time will be set to the most current time in any of the currently displayed **PEN FILES**.

### Changing The Graph Start Time And Date By Using PAN

**PAN** is used by moving the cursor to a desired **Start Time** on the graph then depressing the **PAN LEFT** or **RIGHT** Function Key. This causes the graph to be redrawn with the graph Start Time and Date of wherever the cursor was.

### Changing The Graph Period

The **Graph Period** is changed by tabbing to the **Period Field** and depressing the **CHANGE POINT** Key. At this time, the desired period will be requested. The period is entered as **"1D"**, **"1H"**, or **"1.02H"**, etc.

**Changing The Graph Period By Using ZOOM**

**ZOOM** is used by moving the cursor to a desired **End Time** on the graph then depressing the **ZOOM** Function Key. This causes the graph to be redrawn with the **Graph Period** being the time between the right axis and the cursor (the **Start Time** is unchanged).

**DRAW LATEST Function**

This Function will check each Pen which is currently defined and determine the time of the freshest data available and set the **Graph Start Time** equal to this. This Function is typically used to update the Display with any new data which has been logged to the file since the last **DRAW LATEST**.

**UPDATE VALUE Function**

**UPDATE VALUE** is used to obtain the value of the Pen at the current cursor location. Move cursor to the desired time on the graph and depress the **UPDATE VALUE** Function Key. The values at the cursor of each Pen will be displayed at the bottom of the Screen below the Pens.

**LEFT And RIGHT AXIS Function**

The **LEFT AXIS** Function Key causes the labels on the **LEFT AXIS** of the graph to toggle through the **LEFT AXIS** Labels for each Pen. The **RIGHT AXIS** Function Key does the same for the **RIGHT AXIS**.

**SAVE SETUP Function**

This function will prompt for a filename (use no extension). The current setup will then be saved to the file "FILENAME.PST".

**LOAD SETUP function**

This Function will prompt for a **FILENAME** (use no extension). The previously saved setup will then be loaded from the file **"FILENAME.PST"**.

**Pen ON/OFF Function**

This Function is used by tabbing to the desired Pen and depressing the **PEN ON/OFF** Function Key. The tabbed Pen will either be turned **OFF** if it was **ON**, or **ON** if it was **OFF**. This Function allows Pens to be easily temporarily removed from the graph and then added back again. A Pen which is **OFF** has its text displayed in **gray**.

**PRINT Function**

The **PRINT** Function will print the current display to the default printer for the **ANALYSIS CONSOLE (CONSOLE 2)**. The type of printer and various print options are specified in the file **ALARMCON.IBM**. Currently **Trendgraph** supports black and white printouts to an Epson compatable graphic printer. Color printouts are possible if using a Hewlett Packard PaintJet printer. The following options can be selected from this Screen:

**Printer Type:** Epson, HP Paintjet
**Paper Type:** Paper, Transparency (for Paintjet only)
**Formfeeds:** None, Before, After, Before and After

**RETURN ECMS Function**
This Function will switch consoles to the **ECMS Main Console.**

**Operation Tips:**

To **"Blow Up"** An Area On The Graph.

1. Cursor to right edge of area
2. **PAN**
3. Cursor to left edge of area
4. **ZOOM**

To **Autoscale Pen Limits**

1. **TAB** to desired pen name
2. **CHANGE POINT**
3. **ENTER**
4. **ENTER**
5. **ENTER**

# 8. MAINTENANCE SUPPORT

## MAINTENANCE LOG

The **MAINTENANCE LOG** is called up from the **MORE** Function Key level available on the **SYSTEM Screen** and each of the Drive Screens. This Log can be considered like a **"Cardfile"** where each Log entry is a **"Card"** in the file. The Log Cards can be added, editted, searched for, and printed.

### VIEWING THE LOG CARDS

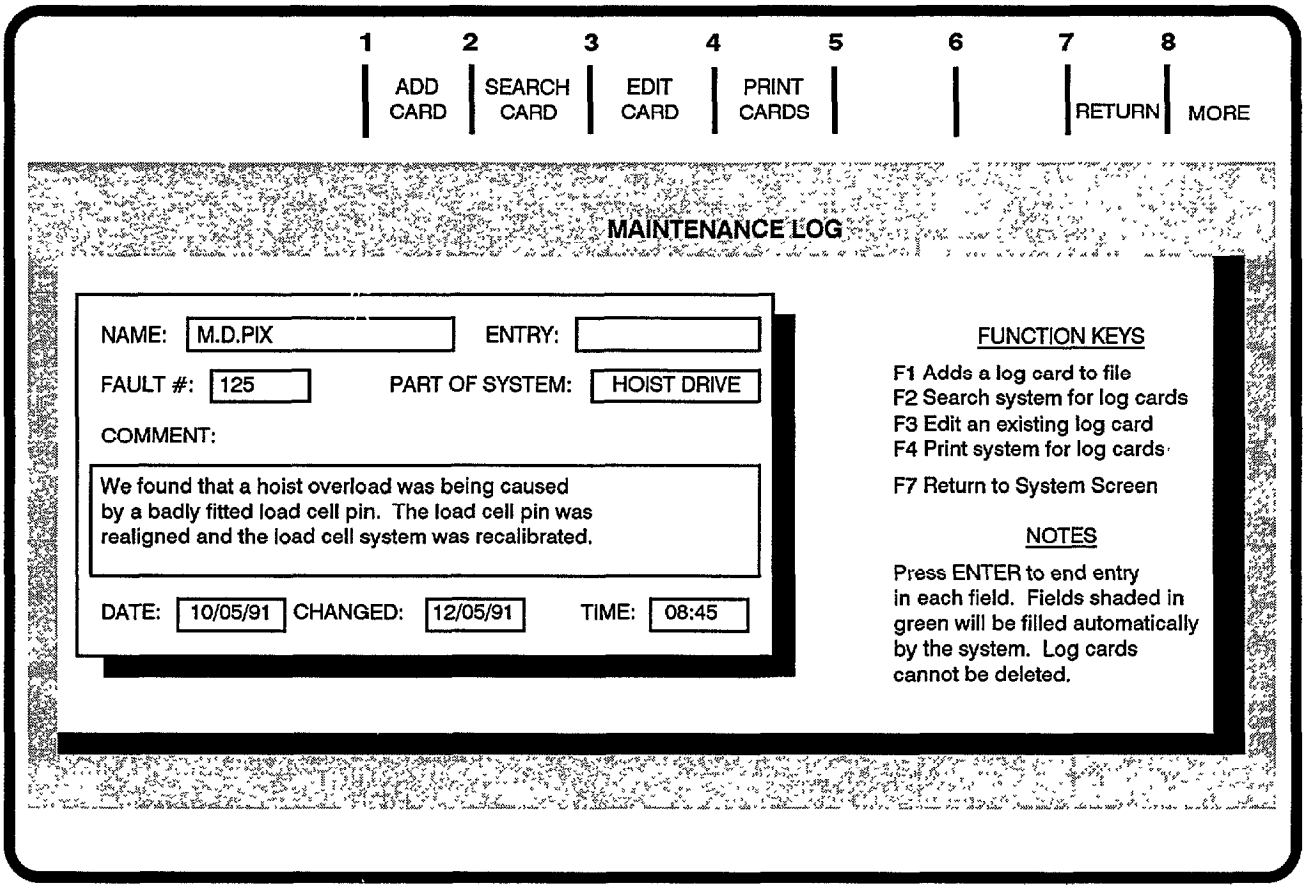
To move through the Maintenance Log Cards, use the following keys:

**HOME -** Moves to first card in log
**END -** Moves to last card in log
**PAGE UP-** Moves one card closer to the begining of log
**PAGE DOWN -** Moves one card closer to the end of the log (or to first blank card if already at the end of log).

### ADD LOG CARD

To get to a blank card, press the **"END"** Key on the keyboard. This will display the last entered Card in the log. Press **"PAGE DOWN"** Key on keyboard. This will cause a new **"Blank Card"** to be displayed. Enter the information on the Card.

**NOTE:** *The **"NAME"** field cannot be left blank - without a name, the card will not be stored. It is suggested that a "Name Convention" is followed to make it easier to search for data later on - an example would be to have each person use their initials. Note that access to the green fields is not possible. These fields are filled in automatically by the log. Each time a new card is created, the date and time are recorded on the card.*



### EDIT A LOG CARD

To edit a Card (one that already exists with data on it), first find the Card required. Using the Cursor **ARROW** Keys and/or the **ENTER** Key, move the cursor to the data to be changed. Modify the data. Next, press the **EDIT LOG CARD** Function Key. The Card will be stored and the **"CHANGE DATE"** and **"TIME"** will be updated to the current Date and Time.

**NOTE:** *Cards cannot be deleted. If a card no longer applies, edit out the information and replace with the words deleted. This method is used to keep the log modifications available.*

### SEARCH FOR CARD

Cards can be searched for by Person's Name, Fault Number, System, and Date. The search process only works from the current Card onwards to the end of the Log. Therefore, if the user wants to search the whole log, press the **HOME** Key first. Pressing the **SEARCH** Function Key will prompt the user to enter the **"Search Criterion"**. Note that the program will only find Cards that match exactly. All spaces and words must be identical thus the importance of having **"Standardized" Naming Conventions** and entry methods is important. Once a Card is located, the search can be repeated by again pressing the **SEARCH** Function Key and following the prompt.

### PRINTING THE LOG

Pressing the **PRINT** Function Key will prompt the user with a selection similar to the **SEARCH** Function. The **"ALL"** selection will print the entire Log and should only really be done if the crane is off-line as it may tie up the printer for some time.

## DRIVE PROGRAMMING

Each Drive's Program can be modified, loaded, stored, and documented by the **EECONSOLE** within **ECMS**. The program is activated by the appropriate Function Key on the **MORE** level of Function Keys of the **SYSTEM SCREEN** or **DRIVE SCREENS**. Once running, **EECONSOLE** offers a number of options (available by Function Key).

### READ EE

This Function allows the user to read 35 consecutive **EEROM** locations from a single LAN drop starting at a user specified address. The data presented will include (for each value) **the Address** in decimal, the **Mneumonic** for the value, the **Value in unsigned decimal**, and **the value in hexadecimal.** These values are stored in the memory of **the ECMS**, and the user may now alter the value (in the **ECMS**) by entering a new value in unsigned decimal or hexadecimal in the **"NEW VALUE"** column of the Display by tabbing to the Parameter to be modified.

### WRITE EE

This Function allows the user to write back to **EEROM** the values entered as above. This Function, prior to writing, checks to see if the **Software Protection** is disabled and if the hardware jumper is in the correct position. It also sets the most significant bit of **EE251** to force the LAN to echo back the **EEROM** value changed for verification. This function allows the writing of **EEROM** as long as power is on the drive and the hardware protection jumper is in the correct position.

**WARNING - MECHANICAL MOTION HAZARD: It is up to the user to insure that a running drive will not be affected. Only values that are changed are written.**

**EE<DISK**

This Function will allow all of the LAN drops **EEROM** values to be stored to disk. The function also has a batch **"Archive" Mode** that saves all the **EEROM** values from all the drives currently listening on the LAN unattended. The user is asked if they require a **Batch Archive** or **Single Drive Recording**. The user must then name the **EEROM** Disk File. For Batch Archives, each drive will have its own file. The user (in this case) is allowed to enter an **8 Character Name**, and the file name will be appended with an extension relating to the drop number from which the **EEROM** was recorded. For example, the user enters **EEDRV**, and the full name will be **EEDRV.###** where ### is the drop number of each drop recorded.

**DISK>EE**

This Function allows a Disk File of **EEROM** values to be written back to a drop. Unlike **EE<DISK**, there is no **Batch File Mode** for this Function. Each file must be written individually. Again, checks are made prior to writing to test the ability to write to the **EEROM**, and all values written will be verified. The Function first compares the present value **EEROM** to the disk value and only writes if they are different.

**COMPARE EE**

This Function allows the user to compare the file of **EEROM** values to any drop on the LAN. The difference can be printed to the **System Printer** or sent to a **Disk File (MISCOMP.DAT)** or both. The printout and file will list the time of the compare, the drop, the file being compared, the address, mnemonic, and values of the miscompared location in the file and the drop.

**EDIT EE**

This Function allows the editing of an **EEROM Disk File** in a similar fashion as the **READ EE** and **WRITE EE** Functions provide. Values can be changed in the Disk File, and the time and date the file was edited will be stored as part of the **EEROM** file along with the time and date the file was originally created.

**PRINT EE**

Under the **EDIT EE Menu**, this Function allows the printout of the **EEROM** values in an **EEROM Disk File**. Header information includes the time and date of the original saving of the file along with the time and date the file was last edited. **EEROM Values** are shown in **unsigned decimal** with their **Address in decimal** and their **Mnemonic.**

**BROADCAST READ**

This Function allows the reading of a **Single Address** from all drops listening on the LAN. The user will be asked for the Address to read. Once entered, the **SCREEN** Display will show all the expected drops on the LAN to respond to this command. The drop numbers will be listed followed by either a **"DROP OFF"** or by the Address and Value. If **"DROP OFF"** appears next to a drop number on this Display, it means that at the time the message was sent to read the **EEROM** to the drives, that particular drop was not on the LAN. The drives that are able to respond to this command will first show the Address and then the Value in that address. Both the **Address** and the **Value** will be shown in **unsigned decimal.**

**BROADCAST WRITE**

This Function allows the user to broadcast a value to a **Single Address** in all drops listening on the LAN. The user will be asked the Address he desires to write in all available drops. The Screen shows all the expected drop numbers followed by the status of the write (**NOT ENABLED** - drop echoed wrong value, **DROP OFF** - drop not listening, **NOT VERIFIED** - drop did not echo value). If the **BROADCAST WRITE** was successful and verified, the Display will show the **Address in decimal** and the **Value written in unsigned decimal.** This is the same display format showing the value when **BROADCAST** is read from the drops.

**EE JUMPERS**

This Program is similar to the **EEROM CONSOLE** except that it will present the **EEROM Values in binary as well as decimal.** The only way to edit these values is to toggle the state of a bit within the binary representation of the value. When the user selects this Function, a new Function Key menu will be displayed. The user may then read up to **36** values from the **EEROM** of a single drive. This is done by selecting the **Standard Jumper Display** that shows the commonly used addresses that will benefit by the binary representation of the numbers. The user may also select a **Custom Jumper Display** that he can configure to include up to **36** of any addresses available in the drive. When the **Custom Jumper List** is selected and a **Custom List** already exists, the Program will ask whether to create a new list or use the previously configured list. If the user selects to configure a new list, the Program then instructs on how to do so.

Once the **Jumper List** is displayed, the user may select which **Address** to edit by using the Cursor **ARROW** Keys. The first bit of each word as they are selected will be highlighted. Using the **TAB** and/or **SHIFT TAB** Keys, the user may select the particular bit in the word. As the user **"TABS"** from bit to bit, the decimal weight of the bit and the bit number currently **"TABBED"** is displayed on the bottom right of the **SCREEN** Display. The user may use the **TOGGLE BIT** Function Key to set or reset the selected bit. The decimal representation of the word will be updated as he toggles the bits. The user, by toggling bits in the jumpers displayed, changes the value in **ECMS** memory. To write the jumpers that have been changed, press the **WRITE EE** Function Key.

**NOTE:** *The drive to which the new values are being written must **NOT** be in a running state or an error message will be displayed. Again, as in the other tools, the drive will be checked to insure that the EEROM is enabled for writing before the values are actually written to the drive.*

## SERIES SIX PROGRAMMING

**Logicmaster** (software used to program and document the Series Six Programmable Controller) will not operate concurrently with **ECMS**. Therefore, the user will have to stop the **ECMS** computer, and start the **Logicmaster** Program.

**Start the Logicmaster as follows:**

1. **RESET** the **ECMS** computer.
2. When asked **"Do you want Concurrent DOS?"** ... answer **"N"**.
3. Logicmaster will start automatically
4. Follow the **Logicmaster** instructions from the **Logicmaster Instruction Book.**

Once finished with **Logicmaster**, the user needs to restart **ECMS** in order to re-establish monitoring and data logging.

**NOTE:** *While **ECMS** software is stopped, no alarms and logging is done.*

**To restart ECMS, RESET** the **ECMS** computer. **ECMS** will automatically start if no answer is supplied to the query **"Do you want Concurrent DOS?"** after a few seconds. (**ECMS** takes approximately 45 seconds to restart completely.)

# 9. CRANE MANAGEMENT

## CRANE UTILIZATION DATA

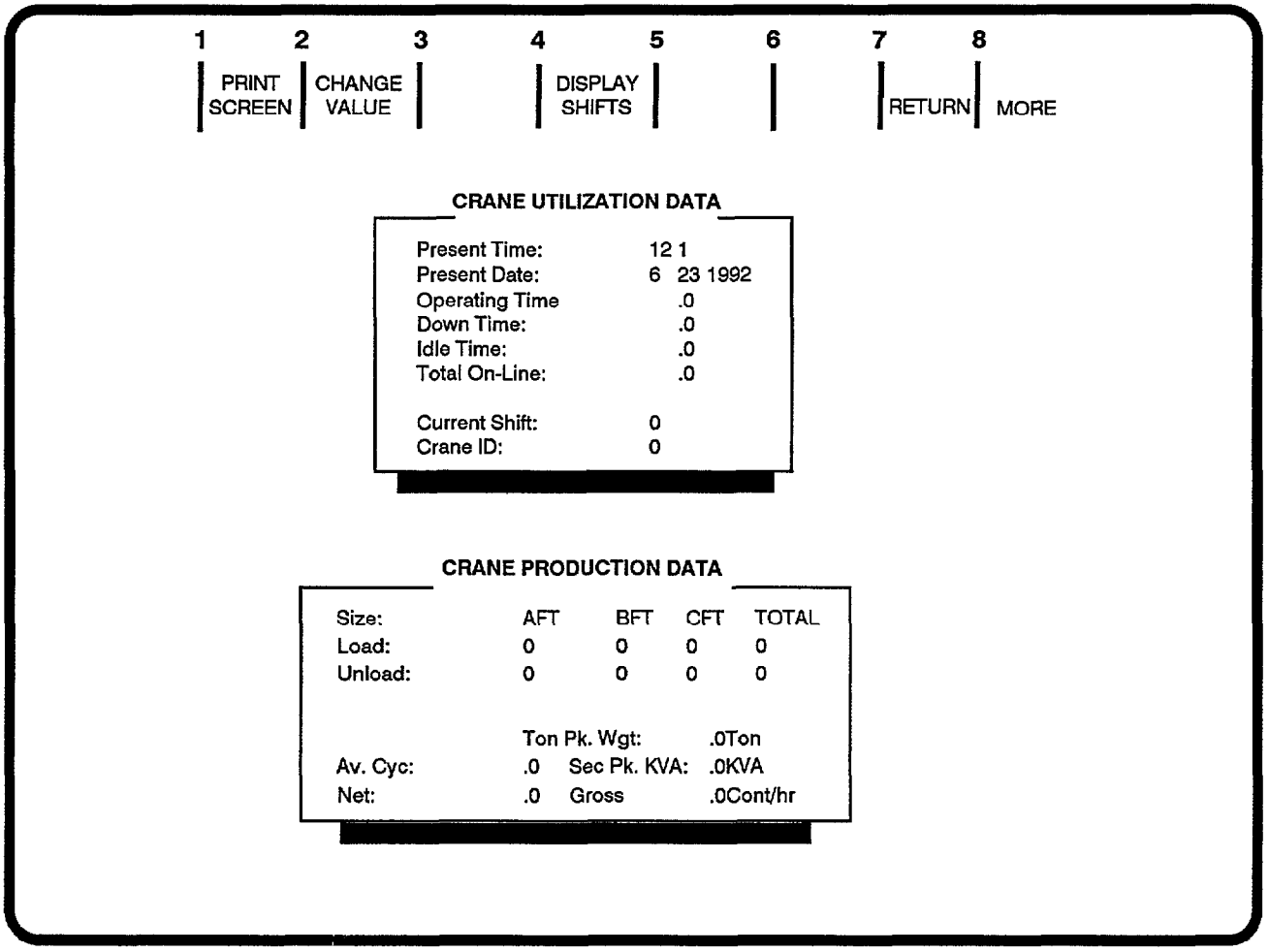
### HOW DATA IS STORED:

Data stored on a **"Per Shift"** basis. Total shifts stored are **140** and each shift is stored by Date and Number (for recall purposes). **Summary Information** is a number of totals spanning the entire **140** shifts, or the number of shifts currently contained in storage. Shifts are numbered (the number that appears on the **CRANE MANAGEMENT** Screen) from **1** to **6** for each day, and are incremented based on the time of day. The shift starting times are shown on the **DISPLAY SHIFTS** Screen and are set from this **DISPLAY SHIFTS** Screen. Simply use the **TAB** Key and **CHANGE VALUE** Key to set the starting shift times to the desired value. Be sure to toggle the **RUN/STOP** Key on the Series Six for the shift times to be read.

### DISPLAYING DATA:

**Two** types of data can be displayed. **Active** (current) data is displayed on the **CRANE MANAGEMENT** Screen. A separate screen, the **DISPLAY SHIFTS** Screen, is designed for displaying **historical** data. This separate screen can display any one of the previous **140** shifts of data, as well as the Summary Information.

To view data from a previous shift, the user enters the **Date** and **Shift Number** into the **ENTRY** Area on the **DISPLAY SHIFTS** Screen. This is achieved by using the **TAB** Key to move the cursor to the correct data field, then the **CHANGE VALUE** Function Key. When a valid **Date** and **Shift Number** has been entered, the user presses the **RECALL SHIFT** Function Key to retrieve the data.



**REPORTING (PRINTING) DATA:**

This output is a selection from the **DISPLAY SHIFTS** Screen. The user is able to print **four** types of reports:

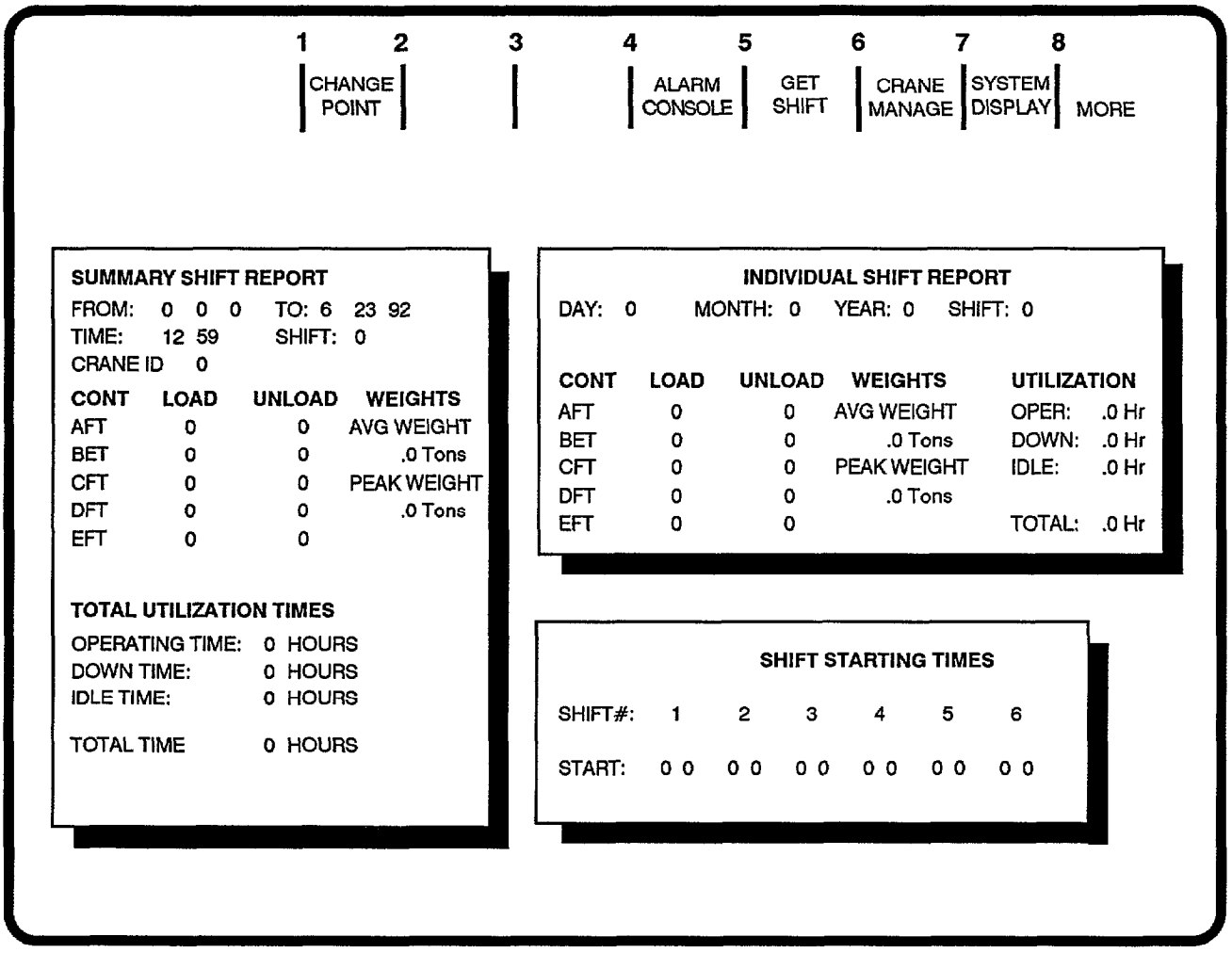
**Current** Shift Data
**Previous** Shift Data (entered by date and number)
**All** Shifts Data
**Summary** Report Data.

**PRESENT TIME** - Current time from the System Clock .

**OPERATING TIME** - Time in hours that the crane has been operating. This is determined by whether any one of the drives (hoist, trolley, gantry, or boom) is running (Run Relay energized). There is a **5 minute** deadband of time from when no drives are operating to when **OPERATING TIME** Timer is stopped. This timer stops immediately a Fault occurs (and **DOWN TIME** Timer will start).

**DOWN TIME** - Time crane has an unresolved (not reset) Fault. This timer will start as soon as a Fault (other than **E/STOP**) is detected and crane is not operating. This timer will stop as soon as Fault is cleared and no Fault exists.

**IDLE TIME** - This is the time that the crane has power available, is reset and ready, but is not being used. This timer will start as soon as crane is reset, and run until a drive is used, or it will start **5 minutes** after the last drive was used. This timer will also time while an **E/STOP** is pressed.



**TOTAL ON-LINE TIME** - This timer times how long the crane has had power applied. The crane may or may not be reset, or have Faults. It is the time that main AC power was available to the crane.

**CURRENT SHIFT** - The current shift's number **(1,2,3,4,5,6)**.

**PRESENT DATE** - When being **displayed**, it will be the system clock. When being **stored** with shift data, it becomes the date for that shift.

**RESET DATE** - This will be the **PRESENT DATE** stored with the first stored shift data. Consequently, as the earliest shifts are discarded (once storage is full and incrementing) the **RESET DATE** will also increment.

## CRANE PRODUCTION DATA

**CONTAINERS LOADED BY LENGTH:**
Container length is determined within the Series Six Program. Once the length of the spreader is known, the Series Six establishes when a container has been lifted up and put down.

**NOTE:** ***ECMS*** *also distinguishes between loading a container onto the ship and unloading from the ship when used with a* ***Dockside*** *crane. When used with a* ***Gantry*** *crane, loaded means "moved".*

**CONTAINERS UNLOADED BY LENGTH:**
Same as for loading containers.

**AVERAGE WEIGHT:**
Weight of each container is available within the Series Six (if the user has a load cell system). A Series Six routine waits until the reading of weight is stabilized (ie. does not include acceleration or snagging). Average container weight is calculated by adding each weight and dividing by the number of containers lifted (both loaded and unloaded) on a per shift basis. For summary, average weights for all shifts are added and divided by the total number of shifts. Peak weight is that weight that is not exceeded by any other for that shift. For summary purpose - the same operation is performed but using all shift peak weights.

**AVERAGE CYCLE:**
This is the time in seconds between the spreader passing over the sill beam towards the ship, and the next time the spreader passes over the sill beam towards the ship. If the cycle exceeds **200 seconds**, it is discarded as not valid. Cycle times are added and divided by total number of cycles counted. For summary, average cycle time will be average of all shift cycle times.

**PEAK KVA:**
The Series Six calculates **KVA** (Volts x Amps x 1.732) each scan and keeps a running peak in a register. This register is reset to zero at the end of each shift. For summary, peak KVA will be that found from all shift peak KVA's This is applicable when user has instrument transformers for voltage and current.

**NET CONT/HR:**
This is the **Net Container Rate Per Hour** achieved by the crane. This is calculated by the total number of containers lifted (loaded and unloaded) divided by the **"Operating Time"**. For summary, overall total number of containers lifted by all shifts is divided by total **"Operating Time"** from all shifts.

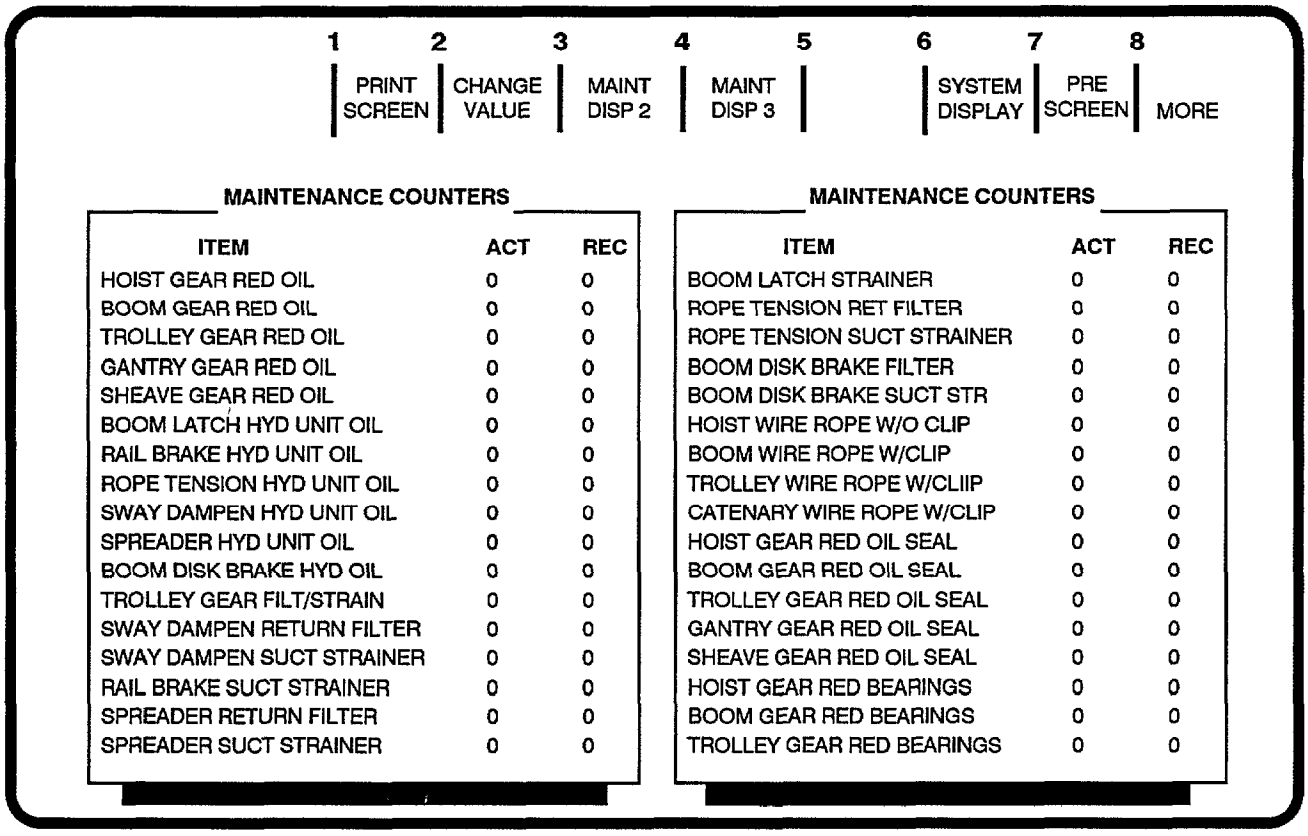
**GROSS CONT/HR:**
This is the **Gross Container Rate Per Hour** achieved by the crane. Calculated by total number of containers lifted divided by length of current shift . For summary, calculated by total containers for all shifts divided by total **On-Line Time.**

## CRANE MAINTENANCE DATA

There are three **CRANE MAINTENANCE** Screens available in **ECMS.** Each Screen can monitor **34** crane-related items for a total of **102** points. Typically, this Function is set up in the field. For each **Crane Maintenance Item** to be monitored, there is a counter in the Master Controller (Series Six). The master controller has **15 timers, set for 1 minute each,** that can be used for this Maintenance Function. Each **Maintenance Item** that has maintenance performed on a time basis would have an interlock from the device energizing a **Maintenance Minute Timer.** The output of the **Maintenance Minute Timer** would drive the appropriate **Maintenance Counter.** Each maintenance item that has Maintenance performed on an **EVENT** basis would have an interlock from the **EVENT** driving the appropriate **Maintenance Counter.**

Each **Maintenance Item** has a **Maintenance Counter** driving it. Each **MAINTENANCE SCREEN** has two columns of information, a **Recommended Maintenance Time (REC)** and an **Actual Time (ACT)** of operation. Using the **TAB** Key, you first come to a **yellow** status "bullet". This is the reset of the individual **Maintenance Counter.** Using the **TAB** Key again, you come to the recommended time of the **Maintenance Item.** This is the preset of the individual **Maintenance Counter.** The **Actual Time** of the **Maintenance Counter** is the actual accumulated time of the individual **Maintenance Counter.**

Each **Maintenance Counter** and **Timer** must get set-up. If the proper information is communicated to the factory this Function may be set-up already. The description of each **Maintenance Counter** must also be written on the various **MAINTENANCE SCREENS.** Again, if the proper information is communicated to the factory this description may already be present. The **MAINTENANCE SCREENS** are accessed through Function Key **F7** from the **SYSTEM SCREEN** and the drive motion **MONITORING SCREENS.**



1 PRINT SCREEN | 2 CHANGE VALUE | 3 MAINT DISP 2 | 4 MAINT DISP 3 | 5 | 6 SYSTEM DISPLAY | 7 PRE SCREEN | 8 MORE

**MAINTENANCE COUNTERS**

| ITEM | ACT | REC |
|---|---|---|
| HOIST GEAR RED OIL | 0 | 0 |
| BOOM GEAR RED OIL | 0 | 0 |
| TROLLEY GEAR RED OIL | 0 | 0 |
| GANTRY GEAR RED OIL | 0 | 0 |
| SHEAVE GEAR RED OIL | 0 | 0 |
| BOOM LATCH HYD UNIT OIL | 0 | 0 |
| RAIL BRAKE HYD UNIT OIL | 0 | 0 |
| ROPE TENSION HYD UNIT OIL | 0 | 0 |
| SWAY DAMPEN HYD UNIT OIL | 0 | 0 |
| SPREADER HYD UNIT OIL | 0 | 0 |
| BOOM DISK BRAKE HYD OIL | 0 | 0 |
| TROLLEY GEAR FILT/STRAIN | 0 | 0 |
| SWAY DAMPEN RETURN FILTER | 0 | 0 |
| SWAY DAMPEN SUCT STRAINER | 0 | 0 |
| RAIL BRAKE SUCT STRAINER | 0 | 0 |
| SPREADER RETURN FILTER | 0 | 0 |
| SPREADER SUCT STRAINER | 0 | 0 |

**MAINTENANCE COUNTERS**

| ITEM | ACT | REC |
|---|---|---|
| BOOM LATCH STRAINER | 0 | 0 |
| ROPE TENSION RET FILTER | 0 | 0 |
| ROPE TENSION SUCT STRAINER | 0 | 0 |
| BOOM DISK BRAKE FILTER | 0 | 0 |
| BOOM DISK BRAKE SUCT STR | 0 | 0 |
| HOIST WIRE ROPE W/O CLIP | 0 | 0 |
| BOOM WIRE ROPE W/CLIP | 0 | 0 |
| TROLLEY WIRE ROPE W/CLIIP | 0 | 0 |
| CATENARY WIRE ROPE W/CLIP | 0 | 0 |
| HOIST GEAR RED OIL SEAL | 0 | 0 |
| BOOM GEAR RED OIL SEAL | 0 | 0 |
| TROLLEY GEAR RED OIL SEAL | 0 | 0 |
| GANTRY GEAR RED OIL SEAL | 0 | 0 |
| SHEAVE GEAR RED OIL SEAL | 0 | 0 |
| HOIST GEAR RED BEARINGS | 0 | 0 |
| BOOM GEAR RED BEARINGS | 0 | 0 |
| TROLLEY GEAR RED BEARINGS | 0 | 0 |

## DOWNLOADING CRANE MANAGEMENT DATA TO AN ASCII FILE FOR DATA RETRIVAL

The crane management information data can be downloaded to an **ASCII File** for importation to utility programs such as Lotus, Supercalc, etc. This is accomplished from the **RUNMENU.** Function Key **F3**, from the **RUNMENU**, will convert the **ECMS** data file to **ASCII** format. Function Key **F4** from the **RUNMENU**, will copy the **ASCII File** to Disk Drive A. This file can then be imported into most Utility Programs for analysis and compilation.

# 10. ECMS RECOVERY AND SETUP

As part of the **Automatic Bootstrap Program**, the **License Protection Key** is checked before any of the **ECMS** Programs can execute. If a printer is connected to the Key, the printer **must** be at least powered **ON**. If **no** printer is installed, **ECMS** will eventually give a **SYSTEM ERROR** Message on the Monitor:

**PRINTER NOT READY ... Retry or Ignore**

**NOTE:** *At this time, the user must press the "I" Key on the keyboard to continue.*

The communication program in **ECMS** depends on the correct connection of the parallel interface to the Series Six Programmable Controller (the Series Six must be in the RUN mode). If these conditions do not exist, the **Automatic Bootstrap Program** will halt execution and display that no communication was established. The communication will continuously try to become established unless aborted by the user, or the communication eventually becomes established. If it is aborted, the **ECMS** Programs will appear to be started; however, no active data will be communicated from the **ECMS** to the rest of the system. The user must reset the computer (most computers have a **RESET** Button on the front panel of the computer) or cycle its power to begin the **Automatic Bootstrap** again.

The instructions for the hardware and software preparation for the **ECMS** System utilizing the Texas Microsystems Computer System are included in **APPENDIX A.**

## ECMS RECOVERY STRATEGIES

If it is necessary to reload the **ECMS** software due to a computer hard disk Fault, all the procedures in **APPENDIX A** need to be completed. Before starting from the beginning, it is worthwhile to establish the **"Level Of Damage"** within the system. If, for example, the hard disk is still relatively intact, it may not be necessary to run some of the set-up procedures.

## ASSESSING THE DMAGE LEVEL

**Level 1** - Minimum Damage
The computer starts (without a disk in Drive A:), asks the question about **"LOADING CONCURRENT DOS"**, and starts **CONCURRENT DOS**. However, **ECMS** does not start properly.

**Level 2** - Moderate Damage
The computer starts (without a disk in Drive A:), but does not ask about **"LOADING CONCURRENT DOS"**.

**Level 3** - Considerable Damage
The computer will not start without **"CDOS"** (Disk 1 of Installlation Kit) disk in Drive A:.

**Level 4** - Extreme Damage
The computer will not start even with **"CDOS"** disk in Drive A:.

After assessing the **Level Of Damage** from the above list (use the next higher **Damage Level** if your symptoms seem to be in between), proceed to reinstall the software as follows (refer to **APPENDIX A**):

**Level 1** - Proceed directly to **Stage III** (bypass **Stages I & II** completely)
**Level 2** - Proceed to **Stage II, Paragraph C** (bypass **Paragraphs A & B**), then to **Stage III.**
**Level 3** - Perform all **Stage II & III** procedures.
**Level 4** - Perform **Stage I, II, & III** procedures. If unable to perform **Stage II & III**, you have hardware malfunction. (Verify that computer operates after **SETUP** is performed).

# *NOTES:*

# APPENDIX A: ECMS SOFTWARE INSTALLATION

The purpose of this Appendix is to provide the information required to install the **General Electric ECMS Software Package** on the computer. This document should be used with the **Installation Kit** of three (3) high density floppy diskettes.

Prior to installing the **ECMS Software** the existing software already installed on the hard disk in all partitions **must** be backed up if possible. This installation procedure will erase all existing software on the disk. The **ECMS Software System** utilizes the **MS-DOS** Operating System partition in its standard configuration and runs the **Concurrent DOS 386** Operating System.

## I. HARDWARE SET-UP

In order for the software installation to proceed without errors, the hardware **must** be configured correctly. Failure to do so may result in this installation procedure failing to work correctly.

**The computer should have the following hardware:**

1. Math co-processor installed in empty chip socket on CPU Board.
2. At least one high density floppy diskette (**1.2MB** or **1.44MB**).
3. EGA/VGA Video Board.
4. One **40MB** Hard Disk - Type 47.
5. **4MB** of RAM
7. Install the **GE-Fanuc WSIA Board.** The Dip Switches on the Terminator Board should be set to terminate the I/O Bus. This is **not** the normal factory configuration as the switches should all be in the closed position. Refer to the **GEK Data Sheet** supplied with the **WSIA** hardware to verify the Dip Switch settings.

   **NOTE:** *When an I/O cable longer than 50 feet is to be used, the use of a I/O Transmitter Board in the Series Six is required.*

## II. HARD DISK PREPARATION

After configuring the memory hardware and running the **SETUP** Program, the hard disk must be properly prepared to accept the **CONCURRENT DOS** Operating System and the **ECMS** Operating Programs.

The hard disk must be organized into **two** logical partitions using the following procedures.

**NOTE:** *The hard disk will receive a **LOW LEVEL FORMAT** to erase old partition table information and all data on the disk. The **Concurrent DOS FDISK Utility** will be used to create a **PCDOS** partition on the first **512** cylinders and a further **PCDOS** partition on the remaining **512** cylinders.*

### A. Low Level Formatting

**NOTE:** *This procedure takes approximately 30-45 minutes to complete. When it has completed, the hard disk will be completely erased. The disk is now ready for partition creation.*

1. Place the **TMI** supplied Diagnostic Diskette into the floppy drive.

2. **RESET** the computer by pressing the front panel **RESET** Button on the computer or cycle its power.

3. Select **HARD DISK FORMAT** Menu Option and answer the Menu prompts by typing the first character of the commands. Use the following answers when asked:
   -- default interleave factor of (2)
   -- quick verification (3 passes)
   -- format without defects list

### B. PCDOS Partition Creation

**NOTE:** *This procedure takes approximately 30 minutes to complete.*

1. Insert **CDOS Disk** of the **ECMS Installation Kit** (Disk 1 of Installation Kit) into the floppy drive.

2. Press the front panel **RESET** Button or cycle power to "Bootstrap" the **Concurrent DOS** Operating System.
   -- If the operating system fails to Bootstrap correctly the computer hardware has not been configured correctly.
   -- Verify that all the available memory is error free.
   -- Run this **Check Disk Utility** to check for lost hard disk clusters
   -- If lost clusters exist, from the **C:\Root Directory** type **"CHKDSK\F"**. This will convert the lost chains into **CHECK** files. Then type **DEL*.CHK** to delete the **CHECK** files.

3. Run the **FDISK** Program as follows:
   a. Type **FDISK** at the **A>** prompt.
   b. Select the option to **"Create a DOS Partition"**.
      -- When asked if all **40MB** are to be used for this partition, answer **NO.**

4. Enter Starting Cylinder of 1, then enter size of **512** cylinders.
   -- When asked for a volume label use **"PCDOS"**. (This partition will become **Drive C:** under **DOS.**)

### C. Additional PCDOS Partition Creation

1. Repeat the procedure outlined in **Paragraph B** to create a further partition.
   -- Select maximum partition size (around **511** cylinders).

2. After creating the partition, enable **PCDOS Bootstrap** from Main Menu. (This partition will become **Drive D:** under **DOS.**)

## III. SOFTWARE INSTALLATION

1. Insert **CDOS** Disk, (Disk 1) of the **ECMS Installation Kit** into the floppy drive.

2. Press the front panel **RESET** Button or cycle power to **"Bootstrap"** the **Concurrent DOS** Operating System.
   -- If the Operating System fails to bootstrap correctly the computer hardware has not been configured correctly.
   -- Verify that all the available memory is error free.

**NOTE:** *To verify that the hard disk has been prepared properly, attempt to switch to the* ***C:*** *and* ***D:*** *Drives by typing the commands:*

***C:*** *followed by a <cr>*
***D:*** *followed by a <cr>*

*If everything is correct, you should receive a new prompt after each of the above commands. If not, start this procedure again from Step 1.*

3. Place **Disk 1** of the 3 Disk **ECMS Software Installation Kit** into **Drive A**, then **RESET** the computer.

4. Type **INSTALL** when the **A>** prompt appears on the Screen.

5. Change disk to **Disk 2** when prompted.

6. Change disk to **Disk 3** when prompted.

7. Hit any key to permanently load the software when prompted - Do Not hit **CTRL-C**.

After installation is complete, remove the disk from the floppy drive and press the **HARD RESET** Button on the TMI computer. The **Concurrent DOS Operating System will** boot and the **ECMS Software** will now run.

# *NOTES:*

# APPENDIX B: SYSTEM CONFIGURATION (For Information Only)

## CONCURRENT DOS OPERATING SYSTEM

In order to run the **ECMS Software**, the Operating System must be properly configured. The **ECMS Installation Procedure** previously described details how to install the software. The Concurrent DOS Manual describes in more detail the various operating system commands.

**ECMS** is normally configured to use a ram disk labeled **Drive E:**. The **CCONFIG.SYS File** is used to create the ram disk using the TMI provided **FASTDISK.SYS** device driver. The supplied **CCONFIG.SYS File** is normally correct to provide this feature.

Printers connected to the system are given numbers depending on the serial or parallel port used to connect them. The following numbers can be used:

| | |
|---|---|
| **LPT1:** | **0** |
| **LPT2:** | **1** |
| **LPT3:** | **2** |
| **COM1:** | **3** |
| **COM2:** | **4** |

Edit the **STARTUP?.BAT** files to select the default printers used for each virtual console.

Run the **SETUP Command** to configure the serial port baud rates, handshaking protocols, etc. from the root directory. If no serial ports are being used in this application, it is not necessary to configure the serial ports. The default configuration of these ports when first installing the operating system is 9600 baud, 8 bits, no parity, XON/XOFF handshaking on input and output (unless a remote **OIT Console** is configured, in which case the 1st serial port will be configured as a multiuser terminal, 19200 baud, non PC emulation).

After setting the port parameters, save system parameters (i.e. serial port configuration). Make sure that you run **SETUP** from the root directory of **Drive C:**. See the Concurrent DOS Manual for more information.

## VIRTUAL CONSOLE LAYOUT

The physical CRT screen of the computer is actually capable of running four programs. Each program is assigned a **"Virtual Console"**. These four Virtual Consoles share the same physical console - the CRT and Keyboard. **Concurrent DOS** automatically manages which programs can read and write from the physical console at any point in time. Pressing the **CONTROL** Key while simultaneously pressing the Numeric Keypad buttons **1** through **4** will switch the control of the screen to the respective Virtual Console.

The **ECMS Software** dedicates the four Virtual Consoles to a specific application program. The commands in the **STARTUP?.BAT** file(s) are used to start the application programs.

- **Console 1** runs the **ECMS** main program using the **START** menu page of the **ECMS.MNU** File. This menu page invokes the batch file procedure named **START.BAT.** After **START.BAT** completes, it invokes **START1.BAT. START1.BAT** actually runs all of the **ECMS** background and foreground tasks.
- **Console 2** runs the **SC-300.MNU** menu system using the **ANALYSIS** menu page. This console will also run **SCTrends** when Function Key **F1** is selected.
- **Console 3** runs the **EVENTCON** Alarm Console Program.
- **Console 4** runs the **ECMS Maintenance Log** and **EECONSOLE.**

## ECMS CONFIGURATION (GE use only)

In order to configure **ECMS** for a real application, the Configuration System should be used. All configuration data is normally stored in the **PROJECT** subdirectory. Before attempting to configure the **ECMS** System for application specific information, be sure to change to this subdirectory. Virtual Console 4 is usually used to configure the system.

To get the Configuration Menu, type **CONFIG** at the **C:\MMI\PROJECT>** prompt. Refer to the appropriate documentation for information on each of these configuration choices.

## I/O TEMPLATES (GE use only)

The normal I/O Template used by the ECMS System is a program named **DAC.CMD**. This program performs the necessary communications between the Series Six Parallel I/O Bus and the WSIA Board set located in the **TMI** Computer. When the DAC program cannot communicate with the Series Six, it disables the ability to change values in the memory resident database of **ECMS**. In order to allow development of software and testing without a Series Six, a dummy I/O Template is provided as part of the **ECMS** Software. This program is named **NOIO.CMD**.

The **START1.BAT** File is used to run the I/O Template. Edit this file and comment out the **DAC** command with a semi-colon and insert the **NOIO** Command. Restarting **ECMS** will then use the dummy I/O Template.

**NOTE:** ***Do not run both*** *the* ***DAC*** *and the* ***NOIO*** *templates simultaneously. If so, changes intended to go to the Series Six may get lost.*

## BACKUP/ RESTORE UTILITY PROGRAMS

Once a system is up and running, the **MMI/Project Directory** must get "Backed-Pp". Put a formatted disk in Drive A, at the **C: \MMI/Project>** Directory, type **"BACKUP"**. The entire contents of the Project Directory will be saved on the disk in Drive A. This disk will serve as backup. To "restore" the "backed up" files, simply type **"RESTORE"** at the **A:>** prompt of the appropriate system with this "Backup" disk in Drive A.

# *NOTES:*



**GE Drive Systems**

**Preliminary GEK-85739B (7/92)**

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